Fundamental Stock Price Cycles *

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Abstract. News shocks about higher future capital returns can explain stock price booms and subsequent busts in a two-asset, heterogeneous agent New Keynesian model. The portfolio choice between more liquid and less liquid forms of capital is key, as it allows for a time-varying illiquidity premium. The arrival of investment opportunities induces capital-rich households to hold more illiquid capital at a lower premium, in anticipation of higher future returns on it. The anticipated higher consumption risk due to less liquid portfolios increases the value of more liquid assets, like stocks. When capital returns mean-revert, capital-rich households rebalance their portfolios, which increases the illiquidity premium and causes stock prices to fall. Novel evidence from survey data on portfolio choices of capital-wealthy households during stock price boom-bust cycles supports the key mechanism of the model.

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1 Introduction

Return predictability has been observed for stock market indices in many countries and across time (Cochrane, 2011, Kuvshinov, 2022). It is also directly linked to the rationalization of "excess" price-volatility through the channel of expected changes in discount rates (Cochrane, 2017). Within this rational framework, there are two open questions: first, why does the subjective discount factor of the marginal trader on the stock market fluctuate strongly enough to quantitatively explain stock price volatility? Second, why is this subjective discount factor predictably mean-reverting and counter-cyclical, causing stock price boom-bust cycles that correlate with the business cycle, as we observe in the data? In this paper, I propose an answer to both questions, by building an incomplete markets model with aggregate fluctuations. In light of the fact that publicly traded firm shares are more liquid than other forms of owning productive capital, I argue that stock price fluctuations are driven by fluctuations in an illiquidity premium that moves upon changes in investment opportunities for wealthy households.

Returns to illiquid capital are procyclical. The importance of income from illiquid asset ownership for asset pricing arises from the group of wealthy, but liquidity-poor households: so-called "wealthy hand-to-mouth" households (Kaplan et al., 2014). I find that procyclical asset income induces more consumption risk for wealthy households than procyclical wage income does for poorer households. Therefore, capital-rich households become the marginal traders in the market for liquid assets. The relevance of this mechanism for asset pricing is amplified when more households hold illiquid portfolios, which happens endogenously in anticipation of a temporary increase in capital returns. Therefore, news shocks about future investment opportunities play an important part in explaining stock price fluctuations, according to this theory.

To quantitatively assess the mechanism, I build a heterogeneous agent New Keynesian model with portfolio choice that is calibrated with survey data and fitted to U.S. business cycle fluctuations. I model public stocks as a perfectly diversified portfolio of firms' profit shares, which is directly held by households and can be traded each period. While optimal portfolio choice along the liquid-illiquid dimension is crucial, the portfolio duration of the liquid asset is indeterminate in the model, as households do not account for aggregate risk. Hence, all households hold the same long-duration liquid asset portfolio, which consists of public stocks and government bonds. Households have time-separable preferences and are modestly risk-averse. In a simulation exercise, I show that the model accounts for two

thirds of the fluctuations in the price-dividend ratio of the aggregate U.S. stock market. I find that the price fluctuations are driven by discount rate news, i.e., the expectations of movements in future stock returns, only when the trading friction on privately held capital is present. Calibrating parts of the model to the "dot-com" boom of the 1990s, I find that it explains about half of the rise in stock valuation during that time.

The theory can be understood as proposing a time-varying illiquidity premium, rather than a time-varying aggregate risk premium (Campbell and Cochrane, 1999, Bansal and Yaron, 2004), as the main explanation for stock price fluctuations. I define the (ex-ante) illiquidity premium as the (expected) difference between the return on illiquid assets and the return on liquid assets[1]. The illiquidity premium varies due to the time-varying propensity to bear consumption risk at the individual level. The expectation of higher future returns on illiquid assets induces wealthy households to bear more consumption risk, by holding more illiquid assets, in the anticipation phase. Thus, the illiquidity premium is low in the anticipation phase. Once illiquid asset returns mean-revert, the illiquidity premium rises above its steady state value, since the marginal traders have more illiquid portfolios and face falling incomes, so that liquid assets become more valuable. Since public stocks are relatively liquid, compared to other capital assets, the growth in stock prices correlates negatively with the illiquidity premium. In this paper, I analyze only the effect of the illiquidity premium on the stock price cycle, and abstract from an aggregate risk premium[2].

In the model, stocks are claims to a share of the profits of the monopolistically competitive firms in the economy. Stock supply is time-invariant (normalized to one), so that I abstract from financing decisions of firms. Hence, a stock price boom does not ease firms' financing constraints. However, a growing stock market is "indirectly" productive in the model by virtue of investors who use the additional liquid asset income to invest

[1]In the literature, this is also called the liquidity premium, see e.g., Bayer et al. (2019).

[2]I solve the quantitative model up to first order (the news about technological progress is an unexpected "MIT-shock"). I conjecture that solving the model non-linearly would not diminish the role of the time-varying illiquidity premium for explaining rising stock prices: in stock price booms, the share of wealth that is held in stocks rises (mechanically, but also by active stock-investment; Section 5 provides evidence for this). If stocks are risky, this increases the riskiness of households' portfolios, which in turn increases the risk premium households are willing to pay, and puts *downward* pressure on stock prices. This mechanism is well-known for consumption-based asset pricing models with aggregate risk. Chien et al. (2012) generate rising stock price booms with risky stocks by having "Mertonian" investors, who price the asset, *sell* their risky shares to intermittently rebalancing investors during a boom, which circumvents the problem. For the housing boom of the early 2000s, Favilukis et al. (2017) conclude that relaxed financing constraints, that is, an institutional change that makes housing an individually less risky asset, are needed to model a simultaneous house price boom and rising share of housing equity in households' portfolios.

more in productive, illiquid capital[3]. I calibrate productive capital in the model to fixed assets in the U.S. Bureau of Economic Analysis tables. A third of fixed assets is housing. At the same time, Asker et al. (2014) estimate that about half of investment in the non-residential sector is carried out by noncorporate private firms. So-called "pass-through" businesses earn more than half of U.S. business income today, up from about a fifth in 1980, and the owners of these private businesses, rather than the holders of public stocks, make up most of the wealthy in the U.S. (Smith et al., 2019, Zidar and Zwick, 2026). The capital of these firms is not traded on the stock market, and is part of the illiquid, productive capital of my model. In sum, around two thirds of productive capital in the U.S. economy is not financed by stock issuance. A part of the valuation of fixed assets is also due to (unproductive) market power by firms—presumably mostly in the corporate sector—which is passed through exclusively to stock owners in the model.

In Section 2, I demonstrate the mechanism in a highly stylized, but tractable model, showing how capital-wealthy households can be marginal traders of the stock market. In Sections 3 and 4, I explore the mechanism quantitatively in a heterogeneous agent New Keynesian model with two assets. My analysis of stock price cycles in a general equilibrium setting uncovers the dependence of the stock price cycle on the elasticity of liquid asset supply. Government bonds are in less demand once news of investment opportunities arrives. Thus, the fiscal authority faces pressure to reduce its balance sheet. I find that a boom in the anticipation phase only arises when the government obliges, and aggregate liquid asset supply is reduced. Fundamentally, higher real returns are "smoothed out" over the anticipation phase by allowing for the crowding out of less productive liquid assets by more productive capital, so that households' incomes increase long before the productivity boost. The well-known effect of temporarily falling markups in New Keynesian models in response to demand shocks also plays a role in allowing the model to generate a boom (Christiano et al., 2010). In Section 5, I employ the extended Survey of Consumer Finances dataset provided by Kuhn et al. (2020), to which I add the survey waves of 2022 and 2025, to provide evidence that "rentiers"—households whose income mainly stems from illiquid capital that they hold passively—decrease their portfolio liquidity in stock price booms, and increase it in stock price busts, as the model

[3]Since there is a negative borrowing limit, the effect of government bond interest rates on investment is non-monotone—when the rates are expected to grow too high, households who borrow in the liquid asset will have to pay too much on their debt, which deters them from investing in illiquid assets. I revisit this case when discussing the effects of monetary and fiscal policy.

predicts[4].

Related literature. Return predictability has been demonstrated to arise in models imposing preferences with external habit (Campbell and Cochrane, 1999), or recursive preferences with long-run consumption risk (Bansal and Yaron, 2004), among others. A consumption-based explanation of stock prices is in line with the finding by Haddad and Muir (2021) that public equity is most often directly held by households among financial assets, and that its returns are unpredictable by the health of intermediaries. The present paper challenges the "stylized fact" that only time-varying *aggregate* risk premia can explain stock price fluctuations. In fact, I find empirically that future bond returns, in a Campbell and Shiller (1988)-decomposition that also includes excess returns to long-duration bonds, are positively correlated with the dividend yield, once a secular break in the 1980s is controlled for.[5] Kuvshinov (2022) compares the risk factor that causes fluctuations in stock prices with the risk factors that drive fluctuations in returns to housing and corporate bonds. He finds that the risk factors do not comove across asset classes. This finding is inconsistent with theories that hinge on aggregate risk, which affects all those assets, as the main cause of asset price fluctuations. The present theory can accommodate this evidence: stocks differ from housing and corporate bonds in their higher liquidity.[6]

Kekre and Lenel (2022) is a recent contribution explaining stock market fluctuations with heterogeneity in risk aversion. Investment and stock prices comove in response to a monetary policy shock: when it redistributes towards households with lower risk aversion, investment rises, while the risk premium falls. In the present theory, instead, the heterogeneity in the individual riskiness of portfolios arises by means of their liquidity, which is an *endogenous* outcome of optimal portfolio choices in response to idiosyncratic and news shocks. The effectiveness of information and illiquid investment in amplifying booms in HANK models with portfolio choice is highlighted in Auclert et al. (2020).

[4]Specifically, the change in the relative portfolio liquidity of the "rentiers" over subsequently sampled years, and the growth in the stock price-dividend ratio, comove negatively, with a standardized coefficient of -0.29. See Section 5 for a discussion of the evidence.

[5]The dividend yield was high before 1980, while ex-post government bond return news were low due to high inflation in the 1970s, both in comparison to the respective mean values after 1980. See Section 5. Similarly, van Binsbergen (2020) finds that a long-maturity government bond portfolio performed as well as the stock market since the 1980s, calling into question the existence of an equity premium for long-duration, liquid asset portfolios.

[6]Sorensen et al. (2014) find that the illiquidity of private equity is an important component of its return risk. Sagi (2020) finds the same for investments in real estate, and rationalizes the illiquidity of the market within a search and matching model.

Luetticke (2021) shows the importance of the redistribution towards households with a high marginal propensity to invest for the transmission of monetary policy shocks.

As an example of the literature deviating from the "FIRE" (full information rational expectations) framework when explaining asset price fluctuations, Adam and Merkel (2019) develop a model with learning where surprise productivity shocks can trigger an endogenous belief propagation that gives rise to boom-bust cycles in stock prices and investment. Through the lens of the present model, and in contrast to the results in Adam and Merkel (2019) and others, the expectations about future productivity are accurate on average (I discuss noise shocks in Section 5).

2 Illustration of the stock price cycle

In this section, I illustrate the mechanism by which wealthy hand-to-mouth households can drive down the equilibrium return on liquid assets. For the most part, I abstract from portfolio choice between liquid and illiquid assets, and analyze a situation in which all households hold little liquid wealth relative to their income risk, i.e., they are poorly insured, while their illiquid wealth is high. In the full model, this situation applies to a small subset of households, as a result of their portfolio choice, at the end of the anticipation phase. In addition to the technology news, in this simplified setting agents are also subject to a shortage of liquidity in the anticipation phase[7]. I apply the technique by Challe and Ragot (2016) to make heterogeneous agent models with a non-degenerate wealth distribution tractable. I then argue that the main intuition goes through for the general case with poor households in the economy and illiquid portfolio choice.

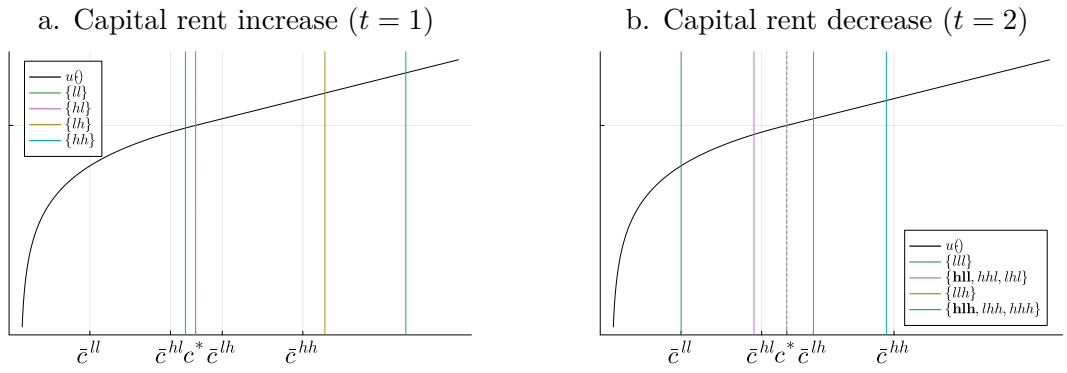
Consider a unit mass of households who hold two assets, a liquid asset and a fixed amount of illiquid capital. They can borrow in the liquid asset up to the constraint $\underline{L} < 0$. Their income consists of the returns on the assets they hold, and idiosyncratic income $y \in \{l, h\}$, $l < h$, which follows a Markov process. They derive utility each period from consumption c, where the utility function $u(\cdot)$ is concave up to the point c^*, and has a constant slope afterwards.

The steady state is calibrated[8] such that all households that receive the low income, l,

[7]This is necessary to bring the market for liquidity into equilibrium: the news about future productivity lowers the demand for liquid savings. Real returns are bounded above by the inverse of the time preference rate, and therefore cannot rise enough to fully offset the lack of demand for liquid assets.

[8]Table A.1 in the appendix shows the values of the parameters.

Figure 1: Optimal consumption levels



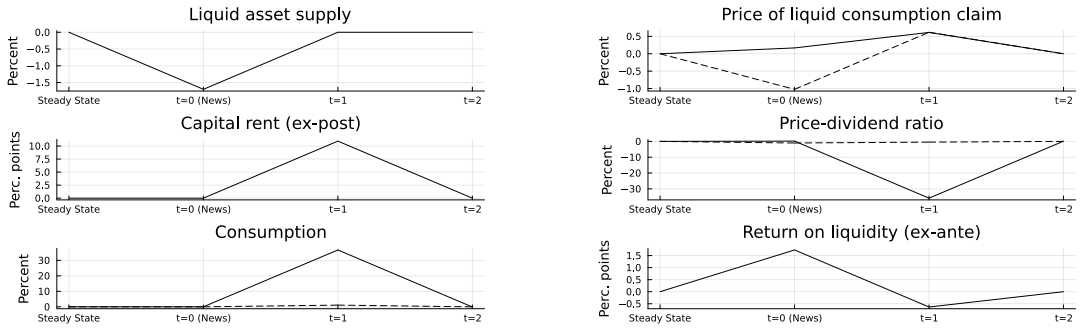
Notes: $\{...y_{t-1}y_t\}$ denotes the history of income shocks at time t, with $y_t \in \{l, h\}$, $l < h$. $\bar{c}^{yy'}$ denote the optimal consumption levels at all possible states $\{yy'\}$ of the ergodic wealth and income distribution.

consume at a level below c^*, which is so low that they would like to borrow more than $\underline{L}$. On the other hand, all households that receive income h consume at a level above c^*. They want to self-insure against the risk of receiving the low income, and hence save $\tilde{b}$ liquid assets. Since they consume at the linear segment of the utility function, their marginal utilities are all identical, so that $\tilde{b}$ is the optimal saving for all households with high income. The economy has a liquid outside asset in positive net supply $L = \pi^l \underline{L} + (1 - \pi^l)\tilde{b}$, where π^l is the unconditional probability of receiving a low income.

The grey lines in Figure 1 show the steady state consumption allocation in the model. Since all households hold the same (positive) amount of fixed capital, the joint distribution over income and liquid asset wealth has only four mass points in steady state: $(l, \underline{L})$, $(l, \tilde{b})$, $(h, \underline{L})$, and $(h, \tilde{b})$. In a first step, I consider a surprise, one-period increase of the capital rent. I choose a rent increase such that households who change from the high to the low state, (hl), now optimally consume c^* and save a positive amount b' for self-insurance. In other words, they become unconstrained due to the higher capital income, but since they face lower capital income again in the future, they want to save part of their income gains. Since the liquid asset supply is constant, the households who receive high income today have to save less than $\tilde{b}$ this period for the bond market to clear. Equilibrium is obtained with a falling return on the liquid asset. For simplicity, I assume the income process to be symmetric[9], so that high-income households will also save the amount $b' < \tilde{b}$. As a result, next period, those households that were lifted out of the constrained state due to the higher capital income are at higher consumption levels than in steady state, while

[9]I choose the conditional probabilities of losing a high income (e.g., job separation) and gaining a high income (e.g., job finding) to sum to 100%.

Figure 2: Impulse responses



Notes: Responses to a news shock about higher future capital rents, and a *simultaneous* surprise drop in the asset supply, at $t = 0$. Dashed lines are for the case with a share of $\alpha = 97\%$ perfectly insured, capital-poor households (see Section 2.2).

households that received high incomes last period consume slightly less (see Figure 1b).

In a second step, I consider the case where the capital rent increase is anticipated one period in advance. To keep the solution tractable, I require that the optimal consumption and liquid asset choices stay the same as above, once capital rents change. This implies that unconstrained households decide to fully insure themselves upon the news (since next period, even if they get low income, they will be unconstrained due to higher capital income). Therefore, the equilibrium return on liquid assets has to increase to $1/\beta - 1$ (β being the time discount factor). For this to be an equilibrium outcome, bond supply has to be depressed in the period of the news shock.

Figure 2 shows the responses of the return on liquid assets (ex-ante), the price of a liquid consumption claim (i.e., the "stock" price), and its price-dividend ratio, to this experiment. The price of the consumption claim appreciates at the onset of the news. It is also higher than in steady state in period $t = 1$, due to the lower liquid asset return then. The price-dividend ratio also increases upon the news. However, the increase in the dividends, once the capital rent rises in the subsequent period, has a larger effect in this calibration. Still, the result illustrates how anticipation can generate a stock price cycle as seen in the data, i.e., high stock prices followed by low returns.

2.1 Equilibrium prices from household optimization

Unlike Challe and Ragot (2016), I consider an equilibrium where the household optimization determines the return on liquid savings endogenously. I abstract from risk in

aggregate variables. For all households i in the economy, it has to hold that

$$u'(c_t^i) \geq \beta R_t \, \mathbb{E}_t^i \left[u'(c_{t+1}^i) \right], \tag{1}$$

where equation (1) holds with equality for all unconstrained households (i.e., households with a high income realization in steady state), and R_t denotes the *ex ante* gross return on liquid savings. In terms of stochastic discount factors $SDF_{t+1}^i := \beta \frac{u'(c_{t+1}^i)}{u'(c_t^i)}$, the equilibrium condition can be written as

$$\frac{1}{R_t} \geq \mathbb{E}_t^i \left[SDF_{t+1}^i \right] \; \forall i. \tag{2}$$

The necessary optimality conditions for a pattern of higher-than-steady-state liquid asset returns in $t = 0$, followed by lower-than-steady-state liquid asset returns in $t = 1$, are thus:

- When liquid asset returns are above steady state, $R_0 > \overline{R}$, it must hold that $\mathbb{E}_0^i \left[SDF_1^i \right] < \overline{ESDF}^i$ *for all* households i[10]
- When liquid asset returns are below steady state, $R_1 < \overline{R}$, there *exists* a household j where $\mathbb{E}_1^j \left[SDF_2^j \right] > \overline{ESDF}^j$. j must be unconstrained by the borrowing limit on liquid savings.

The last condition on household j follows because the level of the liquid asset return is always determined in equilibrium by the households for whom equation (2) holds with equality, i.e., by unconstrained households.

In the example above, both conditions are fulfilled: since in period $t = 1$, households at all wealth and income positions consume more than in steady state (see Figure 1a), all households discount the future by more upon the news in period $t = 0$. In period $t = 1$, there are three unconstrained household types: those with income histories $\{hl\}$, $\{lh\}$, and $\{hh\}$, who all save the amount $b' > \underline{L}$. Since they all have the same expected marginal utility of consumption in period $t = 2$ (under the assumption of the symmetric income process), and the same marginal utility of consumption today (as they consume at the linear segment of the utility function), their expected stochastic discount factor is the

[10] $\overline{ESDF}^i$ denotes the steady state expected stochastic discount factor of household i.

same. It is given by (in terms of households with a high income realization today)

$$\mathbb{E}_1^h \left[SDF_2^h \right] = \frac{\beta}{\gamma} \left(\pi^{hl} u'(c_2^{hl}) + (1 - \pi^{hl})\gamma \right), \tag{3}$$

where $\gamma := u'(c) \; \forall \; c \geq c^*$ is the slope at the linear part of the utility function, and π^{hl} denotes the conditional probability of falling to the low income level from the high income level. The condition $\mathbb{E}_1^h \left[SDF_2^h \right] > \overline{ESDF}^h$ is then equivalent to $c_2^{hl} - \bar{c}^{hl} = R_1 b' - \overline{R}\tilde{b}$ being strictly negative. This is the case, as $R_1 < \overline{R}$ and $b' < \tilde{b}$. Intuitively, the additional income from the illiquid asset holding in period $t = 1$ allows more households to purchase consumption claims for period $t = 2$, which, by goods market clearing, implies that the high-income households expect to consume less, and are therefore willing to save the same amount at lower expected returns.

2.2 Extension: segmented markets

The fluctuation of aggregate consumption in this model economy, where all households hold a large amount of illiquid capital they cannot use to smooth income shocks, while they are close to the borrowing constraint in liquid assets, is two orders of magnitude too large compared to quarterly consumption fluctuations in U.S. data. This is by design: the example was built to illustrate the consumption risk to which these "wealthy hand-to-mouth" households, who then price the liquid asset, are exposed. The idea that incomplete markets can generate realistic price fluctuations, while aggregate consumption remains flat, follows the seminal work by Constantinides and Duffie (1996). In order to clarify this contribution of my paper, I now insert more households into the model economy.

The newly introduced households insure themselves perfectly against idiosyncratic income shocks (trading Arrow securities amongst themselves), but cannot access "outside" financial markets, in the sense that they cannot hold capital, and cannot issue debt to households that hold capital (segmented markets). The optimality condition with respect to their liquid asset holding is then

$$\frac{1}{R_t} \geq E_t^\alpha \left[SDF_{t+1}^\alpha \right] = \beta, \tag{4}$$

where α denotes capital-poor households, who do not have consumption risk and thus have a constant discount factor β. By complementary slackness, their saving in liquid assets is zero if inequality (4) is strict. If they are indifferent (when $1/R_t = \beta$), I assume

that they decide to stay at the borrowing constraint, i.e., $b_t^\alpha = 0$. Note that, since R_t peaks at $1/\beta$ in period $t = 0$ in the above experiment, the optimality condition (4) is always fulfilled, and the liquid asset is still priced by the uninsured households.

Let α denote the share of perfectly insured households that do not hold capital in the economy. Since they have no income besides l or h, they consume the constant $c_t^\alpha = \pi^l l + (1 - \pi^l)h =: \overline{y}$. Households who hold capital, but cannot trade Arrow securities to insure themselves against income shocks, consume $\tilde{c}_t := \sum_{j \in J} \pi^j c_t^j$, where J encompasses all possible income histories $\{lll\}, \{hll\}, ..., \{hhh\}$, and π^j is the probability weight of these histories. The aggregate consumption is then given by $c_t = \alpha\overline{y} + (1 - \alpha)\tilde{c}_t$. Choosing α high enough such that the consumption of insured, capital-poor households makes up more than 90% of aggregate consumption in steady state then yields an attenuation of aggregate consumption fluctuations by almost two orders of magnitude[11], while the fluctuations in the returns to the liquid asset remain unchanged (see the dashed lines in Figure 2). The movements in the price-dividend ratio are attenuated; in the quantitative model, stocks are only claims to a fraction of output, and dividend payments are smoothed out, so that return volatility will have a bigger impact on the price-dividend ratio.

2.3 General case: tradable capital

In the full model that I discuss in Section 3, I do not rely on the assumption of segmented markets. Instead, a share of households with high exposure to capital income risk arises endogenously, through the portfolio choice between liquid and illiquid assets. However, the basic insight from Section 2.1 about the optimality conditions that lie behind the boom-bust cycle in stock prices remains, as the following proposition shows:

Proposition 1. *Let λ be the probability that a household can trade capital k_t in period t. Capital is traded at the price q_t. Let R_t^L be the gross return on liquid assets, and R_t^K the gross return on the illiquid capital. Households assume certainty equivalence with respect to aggregate shocks. Then, it holds that*

$$ILP_t := R_t^K - R_t^L \geq \beta(1 - \lambda)\frac{\mathbb{E}_t^i\left[\gamma_{t+1}^i\right]}{q_t u'(c_t^i)} \tag{5}$$

[11] Choosing $\alpha = 0.97$, which roughly corresponds to the 2.3% of households whose income is dominated by capital income in the full model (see below), yields a peak increase of aggregate consumption of about 1%, a reduction by a factor of 35 from the case without insured households.

for all households i that trade capital in period t, where ILP is the illiquidity premium, and

$$\gamma_t^i := q_t u'(c_t^{i,n}) - \beta \, \mathbb{E}_t^i \, V_{t,k}^{i'}(b_t^{i,n}, k_t^{i,n}) \geq 0 \tag{6}$$

is the shadow price of selling capital in period t, where the household cannot trade capital, i.e., $k_t^{i,n} = k_{t-1}^i$, and chooses consumption $c_t^{i,n}$ and liquid assets $b_t^{i,n}$. $V_k^i(b,k)$ denotes the marginal value of holding capital for household i.

Proof See Appendix B.

Note the similarities between equations (5) and (2): $\beta \frac{\gamma_{t+1}^i}{u'(c_t^i)}$ is similar to a stochastic discount factor, augmented by the future value of trading the illiquid asset. Upon the news of higher future returns on capital, the marginal value of holding capital in the future rises—the shadow price of selling capital falls—which is necessary for the illiquidity premium to fall. As an additional requirement for the premium to fall, the stochastic discount factors of households who do not hold capital have to decline, or be sufficiently low, at that point. This comes about in the full model through an investment-driven boom that raises the labor incomes of households.

Once the higher return on capital, which is temporary, has materialized, capital-wealthy households face an income decline. Hence, their expected shadow price of selling capital rises, since they expect to consume less, and the marginal value of holding capital in the future declines. The existence of these households is sufficient to increase the illiquidity premium above its steady state level. Fluctuations in the illiquidity premium will be mainly driven by fluctuations in the return to liquid assets, R^L, and therefore explain the boom-bust cycle in stock prices.

For the rest of the paper, I solve the response to technology news in a HANK model with portfolio choice, which is calibrated to match micro data on labor income processes and wealth inequality.

3 A HANK model of the stock market

The model economy consists of heterogeneous households, who are subject to idiosyncratic income shocks and stochastic (illiquid) capital market access, a production sector with intermediate goods producers, who hire workers and rent capital, and final goods

producers, who set prices subject to price adjustment costs, and a government sector, where a monetary and a fiscal authority react to business cycle conditions by setting the nominal interest rate and the bond supply according to fixed rules. In the following, I describe each sector individually, before stating the market clearing conditions and giving the definition of the equilibrium of the model[12]. The model is partly calibrated to aggregate data of the U.S. economy from 1954 to 2015, and partly estimated by Bayesian methods (see Bayer et al. (2024)). One period denotes one quarter. $\bar{X}$ denotes the steady state value of variable X, and $\hat{X}$ the relative deviation of X from $\bar{X}$.

3.1 Households

There is a unit mass of ex-ante identical households, indexed by i, who are infinitely lived, discount the future with the factor β, and maximize time-separable utility of the Constant Relative Risk Aversion (CRRA) type, $u(x) = \frac{1}{1-\xi}x^{1-\xi}$, over consumption, c_{it}, and leisure. Each period t, they choose consumption, labor supply n_{it}, future holdings of liquid assets, b_{it+1}, and non-negative illiquid/capital assets, k_{it+1}, subject to their budget constraint, the debt limit $\underline{B}$, and their access to the market for the illiquid asset. Their budget is composed of (after tax) labor income, $w_t h_{it} n_{it}$, profit incomes Π_t^F (final goods firms' rents) and Π_t^U (labor union rents), and asset returns. While w_t denotes the aggregate wage rate, their individual productivity h_{it} is determined stochastically according to

$$h_{it} = \frac{\tilde{h}_{it}}{\int \tilde{h}_{it} di}, \tag{7}$$

$$\tilde{h}_{it} = \begin{cases} \exp(\rho_h \log \tilde{h}_{it-1} + \epsilon_{it}^h) & \text{with probability } 1-\zeta \text{ if } \tilde{h}_{it-1} \neq 0, \\ 1 & \text{with probability } \iota \text{ if } \tilde{h}_{it-1} = 0, \\ 0 & \text{else.} \end{cases}$$

$\tilde{h}$ follows a log-AR(1) process, with $\epsilon_{it}^h \sim \mathcal{N}(0, \sigma_{h,t}^2)$, for the times when the household is a worker. Its volatility moves endogenously in response to aggregate hours: $\sigma_{h,t}^2 = \bar{\sigma}_h^2 \exp(\hat{s}_t)$, $\hat{s}_{t+1} = \rho_s \hat{s}_t + \Sigma_Y \hat{N}_{t+1}$. ζ is the probability of becoming an entrepreneur. Entrepreneurs have no labor income ($h_{it} = 0$), but gain a share of the profits of the final goods firms, Π_t^F, and raise funds by issuing stock (see Section 3.2). With probability ι, they return to being a worker with mean productivity. The average of individual

[12]The model setup, with the exception of the modelling of aggregate shocks and the inclusion of liquid stocks, is the same as in Bayer et al. (2024). This is a shortened version of their exposition.

productivity h is normalized to 1. In addition to their wages, workers also receive a lump-sum share of the labor union rent, Π_t^U. The existence of entrepreneurs solves the problem of the allocation of profits that occurs in HANK models. Additionally, it helps the model to match the highly skewed wealth distribution in the data.

The choice of labor supply is greatly simplified by assuming Greenwood-Hercowitz-Huffman (GHH) preferences. They are represented by subtracting the disutility of work, $G(h_{it}, n_{it})$, from the consumption good *within* the felicity function, i.e., $u(c_{it} - G(h_{it}, n_{it}))$. In this setting, an increase in working hours directly increases the marginal utility of consumption, which offsets the typical consumption-labor tradeoff that arises with separable disutility of labor, namely that more work is only compatible with a smaller consumption level. As a result, optimal labor supply is a function only of the net labor income, independent of consumption[13]. Let $x_{it} = c_{it} - G(h_{it}, n_{it})$ denote the composite demand for consumption and leisure.

Labor income of households is subject to progressive taxation as in Heathcote et al. (2017), i.e., net labor income y_{it} is given by

$$y_{it} = (1 - \tau^L)(w_t h_{it} n_{it})^{1-\tau^P}, \tag{8}$$

where w_t is the aggregate wage rate and τ^L and τ^P are the level and the progressivity of the tax schedule. Assuming that $G(h, n)$ has constant elasticity γ with respect to n, the first-order condition for labor supply yields $G(h_{it}, n_{it}) = y_{it}\frac{1-\tau^P}{1+\gamma}$. Choosing $G(h_{it}, n_{it}) = h_{it}^{1-\tau^P}\frac{n_{it}^{1+\gamma}}{1+\gamma}$ simplifies the problem further, as labor supply then is only a function of the aggregate (after tax) wage rate. This implies that every household works the same number of hours, $n_{it} = N(w_t)$.

Households can have unsecured debt (i.e., negative holdings of the liquid asset) up to the borrowing limit $\underline{B}$[14]. In this case, their payment to the lender consists of the nominal liquid rate, R_t^L, plus a wasted intermediation cost, $\overline{R}$. Each period, a household's chance

[13]Jaimovich and Rebelo (2009) propose a class of preferences that nests both King-Plosser-Rebelo (KPR) and GHH preferences, which was then adopted by Schmitt-Grohé and Uribe (2012) and others in their structural estimation of the impact of news shocks. The reason is that GHH preferences, which shut down the wealth effect on labor supply, are helpful in generating booms from news shocks. Hence, having a preference class where this wealth effect enters as a parameter, which can be estimated, gives news shocks a higher chance to fit the data. Schmitt-Grohé and Uribe (2012), as well as Born and Pfeifer (2014) and Bayer et al. (2024) in models without news shocks, find that preferences close to GHH provide the best fit to the data.

[14]Since all households hold a share of their liquid wealth in stocks, for negative liquid wealth they symmetrically do some of their borrowing in stocks ("short-selling" stocks).

of participating in the market for illiquid assets, and being able to adjust k_{it+1}, is given by the fixed probability λ. This trading friction renders capital illiquid. The capital good's price in period t is q_t. From holding capital, households earn a capital rent r_t. The household's budget constraint is

$$c_{it} + b_{it+1} + q_t k_{it+1} = y_{it} + \mathbb{1}_{h_{it}\neq 0}(1-\tau)\Pi_t^U + \mathbb{1}_{h_{it}=0} y_t^e + (q_t + r_t)k_{it} + \left(\frac{R_t^L}{\pi_t} + \mathbb{1}_{\{b_{it}<0\}}\frac{\overline{R}}{\pi_t}\right) b_{it}, \tag{9}$$

where $\pi_t = \frac{P_t}{P_{t-1}}$ denotes realized gross inflation, τ is the average tax rate (see Section 3.4) and y_t^e denotes the after-tax income of entrepreneurs (see Section 3.2). Households maximize the infinite discounted sum of their utility, choosing (composite) consumption, liquid assets, and, if possible, illiquid capital holdings subject to the budget constraint and the inequalities $k_{it+1} \geq 0$ and $b_{it+1} \geq \underline{B}$.

The individual household's optimization problem can be written recursively as

$$\begin{aligned} V_t^a(b,k,h;\Theta,\mathcal{P},\Omega) &= \max_{k',b_a'}\{u[x(b,b_a',k,k',h)] + \beta\,\mathbb{E}_t\,V_{t+1}(b_a',k',h';\Theta',\mathcal{P}',\Omega')\}, \\ V_t^n(b,k,h;\Theta,\mathcal{P},\Omega) &= \max_{b_n'}\{u[x(b,b_n',k,k,h)] + \beta\,\mathbb{E}_t\,V_{t+1}(b_n',k,h';\Theta',\mathcal{P}',\Omega')\}, \end{aligned} \tag{10}$$

$$\begin{aligned} \mathbb{E}_t\,V_{t+1}(b',k',h;\Theta',\mathcal{P}',\Omega') &= \mathbb{E}_t[\lambda V_{t+1}^a(b',k',h;\Theta',\mathcal{P}',\Omega')] \\ &\quad + \mathbb{E}_t[(1-\lambda)V_{t+1}^n(b',k,h;\Theta',\mathcal{P}',\Omega')], \end{aligned}$$

where Θ stands for the distribution over asset holdings and productivity, $\mathcal{P}$ are equilibrium prices, and Ω denotes an exogenous shock.

3.2 Tradable profit-stocks

Liquid assets consist of government bonds (see Section 3.4) and profit-stocks. Profit-stocks are claims to a share of smoothed profits of final goods firms, Π_t^F (see Section 3.3). The smoothing works through a fixed investment rule: A fraction ξ^Π of excess profits, defined as the deviation from steady-state profits, $\Pi_t^F - \Pi^F$, becomes available for payment to stockholders and the entrepreneurs (who are the owners of the firms). The rest of the excess profits is saved in a common account, if positive, or withdrawn from the account, if negative. The account is invested in government bonds. Its wealth is denoted by NW_t^Π at the end of period t. At times when firms are net borrowers, they do not pay the borrowing

wedge that households pay, and are not subject to a borrowing constraint. On average, the account holds zero wealth, $NW^{\Pi} = 0$. A fraction ξ^{Π} of the interest payments on the wealth held in the account becomes available to stockholders and the entrepreneurs, while the rest is reinvested. The smoothed profits then amount to

$$\tilde{\Pi}_t^F := \xi^{\Pi}(\Pi_t^F + NW_{t-1}^{\Pi} R_t^b/\pi_t) + (1-\xi^{\Pi})\Pi^F \tag{11}$$

A fraction ω^{Π} of the smoothed profits is traded with a unit mass of shares every period at price q_t^{Π}. A fraction ι^{Π} of those shares retires every period and loses its value, while new shares are issued by the entrepreneurs. The real, after-tax payout to entrepreneurs then becomes

$$y_t^e := (1-\tau^L)((1-\omega^{\Pi})\tilde{\Pi}_t^F + \iota^{\Pi} q_t^{\Pi})^{1-\tau^P} \tag{12}$$

Ex-ante, the expected return on bonds, R_{t+1}^B, has to fulfill the no-arbitrage condition

$$\mathbb{E}_t \frac{R_{t+1}^B}{\pi_{t+1}} = \mathbb{E}_t \frac{q_{t+1}^{\Pi}(1-\iota^{\Pi}) + \omega^{\Pi}\tilde{\Pi}_{t+1}^F}{q_t^{\Pi}}. \tag{13}$$

With B_t denoting the total supply of government bonds at time t, the total supply of liquid assets at time t becomes $L_t = B_t + q_{t-1}^{\Pi}$. The average (ex-post) real return on liquid assets is then given by

$$\frac{R_t^L}{\pi_t} = \frac{B_t}{L_t} \cdot \frac{R_t^B}{\pi_t} + \frac{q_t^{\Pi}(1-\iota^{\Pi}) + \omega^{\Pi}\tilde{\Pi}_t^F}{L_t}. \tag{14}$$

3.2.1 Accounting of capital gains

To be in line with the data (see below), I count capital gains as part of wealth gains instead of income. Capital gains can accrue from illiquid capital, $\frac{q_t}{q_{t-1}}$, if households can trade their capital holdings in period t, and liquid stocks, $\frac{q_t^{\Pi}}{q_{t-1}^{\Pi}}$. The budget constraint (9) is already formalized in a way that illiquid capital gains count as wealth gains. For the liquid asset, instead, I introduce the liquid asset *value*

$$q_t^L := 1 + \frac{q_t^{\Pi} - q_{t-1}^{\Pi}}{L_t}. \tag{15}$$

Subtracting q_t^L from the ex-post real return on liquid assets, $\frac{R_t^L}{\pi_t}$, yields the net return on liquid assets (net of capital gains from stocks and stock depreciation):

$$r_t^{L,net} := \frac{R_t^L}{\pi_t} - q_t^L = \frac{B_t}{L_t} \cdot \left(\frac{R_t^B}{\pi_t} - 1 \right) + \frac{\omega^{\Pi} \tilde{\Pi}_t^F - \iota^{\Pi} q_t^{\Pi}}{L_t} \tag{16}$$

The value of liquid assets for a household with liquid saving b_{it} can then be rewritten as

$$\left(\frac{R_t^L}{\pi_t} + \mathbb{1}_{\{b_{it}<0\}} \frac{\overline{R}}{\pi_t} \right) b_{it} = \underbrace{\left(r_t^{L,net} + \mathbb{1}_{\{b_{it}<0\}} \frac{\overline{R}}{\pi_t} \right) b_{it}}_{net\ liquid\ income} + \underbrace{q_t^L b_{it}}_{liquid\ wealth} \tag{17}$$

3.3 Production sector

The production sector of the economy is made up of labor unions and labor packers, intermediate goods producers, final goods firms, and capital goods producers. Workers sell their labor at the nominal rate W_t to a continuum of unions (indexed by j), who sell their variety of labor to labor packers (for W_{jt}), which produce and sell the final labor service at the price W_t^F. Since unions have market power, they set a price $W_{jt} > W_t$ subject to the demand curve $n_{jt} = (W_{jt}/W_t^F)^{-\zeta} N_t$, and to a Calvo-type adjustment friction. In a symmetric equilibrium, their optimization yields the wage Phillips curve (linearized around the steady state)

$$\log\left(\frac{\pi_t^W}{\bar{\pi}_W} \right) = \beta\, \mathbb{E}_t \log\left(\frac{\pi_{t+1}^W}{\bar{\pi}_W} \right) + \tilde{\kappa_w} \left(\frac{w_t}{w_t^F} - \frac{1}{\mu^W} \right), \tag{18}$$

where $\pi_t^W = \frac{W_t^F}{W_{t-1}^F}$ is the gross wage inflation, w_t and w_t^F are the real wages for households and firms, $\frac{1}{\mu^W} = \frac{\zeta-1}{\zeta}$ is the target markdown of wages, and $\tilde{\kappa_w}$ is determined by the probability of wage adjustment, κ_w[15]. The return to the unions is then given as $\Pi_t^U = (1 - \frac{1}{\mu^W}) N_t w_t^F$ in real terms.

The homogeneous intermediate good Y is produced with the constant returns to scale production function

$$Y_t = A_t N_t^{1-\alpha_t} (u_t K_t)^{\alpha_t}, \tag{19}$$

where u_t is capital utilization. As is standard, higher capital utilization implies an increased depreciation of capital, $\delta(u_t) = \delta_0 + \delta_1(u_t - 1) + \frac{\delta_2}{2}(u_t - 1)^2$, where $\delta_1, \delta_2 > 0$. A_t

[15]It holds that $\tilde{\kappa_w} = \zeta \kappa_w \frac{\mu^W - 1}{\mu^W}$.

and α_t are the level of Total Factor Productivity (TFP) and the capital share, respectively, and follow the stochastic processes

$$\log(A_t) = \rho_A \log(A_{t-1}) + \epsilon^{A,\ell}_{t-\ell} + \epsilon^A_t, \tag{20}$$

$$\alpha_t = (1-\rho_\alpha)\overline{\alpha} + \rho_\alpha \alpha_{t-1} + \epsilon^{\alpha,\ell}_{t-\ell} + \epsilon^\alpha_t, \tag{21}$$

$$\epsilon^A_t \sim \mathcal{N}\left(0, \sigma^2_A\right),\ \epsilon^\alpha_t \sim \mathcal{N}\left(0, \sigma^2_\alpha\right).$$

Here, $\epsilon^{A,\ell}_{t-\ell}, \epsilon^{\alpha,\ell}_{t-\ell}$ denote news shocks (technology news, either about TFP or the capital share) that households receive in period $t-\ell$, and which are added to (the logarithm of) the fundamental process ℓ periods later (as indicated by the superscript). ℓ is called the anticipation horizon of the news. In other words, the capital share and log-TFP follow an ARMA process, where the moving average part is known ℓ periods in advance, and hence interpreted as news. This interpretation is standard in the literature (e.g., Schmitt-Grohé and Uribe (2012), Barsky and Sims (2012)). In particular, I assume the news shocks to be i.i.d. draws from the same distribution as the surprise shocks $\epsilon^A_t, \epsilon^\alpha_t$ (i.e., news shocks are not autocorrelated as in Leeper and Walker (2011)).

Let mc_t denote the relative price (compared to the consumption good) at which the intermediate good is sold to final goods firms (which makes it the marginal cost of Y_t for these firms). The intermediate good producers, who operate under perfect competition, set the real wage and the user costs of capital according to the marginal products of labor and capital:

$$w^F_t = (1-\alpha_t) mc_t A_t (u_t K_t / N_t)^{\alpha_t},\ r_t + q_t \delta(u_t) = u_t \alpha_t mc_t A_t (N_t / u_t K_t)^{1-\alpha_t}. \tag{22}$$

Utilization is decided by the owners of the capital goods, who take the aggregate supply of capital services as given, and therefore follow the optimality condition

$$q_t \delta'(u_t) = \alpha_t mc_t A_t (N_t / u_t K_t)^{1-\alpha_t}. \tag{23}$$

Final goods firms (which are owned by the entrepreneurs) differentiate the intermediate good into final goods of the variety j, y_j. In this environment of monopolistic competition, they maximize profits subject to the demand curve $y_{jt} = (p_{jt}/P_t)^{-\eta} Y_t$ and price adjustment frictions. It is assumed that they discount the future at the same rate as the households, β. Then, their optimization yields a symmetric equilibrium that up to

first order is determined by the Phillips curve

$$\log\left(\frac{\pi_t}{\bar{\pi}}\right) = \beta\,\mathbb{E}_t \log\left(\frac{\pi_{t+1}}{\bar{\pi}}\right) + \tilde{\kappa}\left(mc_t - \frac{1}{\mu^Y}\right), \tag{24}$$

where $\mu^Y = \frac{\eta}{\eta-1}$ is the target markup, and $\tilde{\kappa}$ is determined by the probability of price adjustment, κ[16]. The rent of the final goods firms is $\Pi_t^F = Y_t(1 - mc_t)$ in real terms.

Capital producers transform the investment of consumption goods into capital goods, taking as given the price of capital goods, q_t, and investment adjustment costs. They maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t I_t \left\{ q_t \left[1 - \frac{\phi}{2}\left(\log\frac{I_t}{I_{t-1}}\right)^2 \right] - 1 \right\}. \tag{25}$$

Up to first order, the problem reduces to the equation

$$q_t \left[1 - \phi \log\frac{I_t}{I_{t-1}} \right] = 1 - \beta\,\mathbb{E}_t \left[q_{t+1}\phi \log\frac{I_{t+1}}{I_t} \right], \tag{26}$$

which determines q_t from the rates of investment. Since all capital goods producers are symmetric, the law of motion for aggregate capital follows as

$$K_t - (1 - \delta(u_t))K_{t-1} = \left[1 - \frac{\phi}{2}\left(\log\frac{I_t}{I_{t-1}}\right)^2 \right] I_t. \tag{27}$$

3.4 Government sector

In the government sector, a monetary authority (the central bank) controls the nominal interest rate on bonds, while a fiscal authority (the government) issues bonds to finance deficits. The monetary policy follows a Taylor rule with interest rate smoothing:

$$\frac{R_{t+1}^B}{\bar{R}^b} = \left(\frac{R_t^B}{\bar{R}^b}\right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}}\right)^{(1-\rho_R)\theta_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{(1-\rho_R)\theta_Y}. \tag{28}$$

$\theta_\pi, \theta_Y \geq 0$ govern the severity with which the central bank reacts to deviations in inflation and the output gap, where Y_t^* is defined as the output that would be obtained at steady

[16] It holds that $\tilde{\kappa} = \eta\kappa\frac{\mu^Y - 1}{\mu^Y}$.

state markups. The government issues bonds according to the fiscal rule

$$\frac{B_{t+1}}{B_t} = \left(\frac{B_t}{\bar{B}}\right)^{-\gamma_B} \left(\frac{\pi_t}{\bar{\pi}}\right)^{-\gamma_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{-\gamma_Y}. \tag{29}$$

Let $\mathcal{B}_t := \sum_i (w_t n_{it} h_{it} + \mathbb{1}_{h_{it}=0} \Pi_t^F)$ be the tax base for the progressive tax code. The total tax revenue T_t sums up to $T_t = \tau(\mathcal{B}_t + \sum_i \mathbb{1}_{h_{it}\neq 0} \Pi_t^U)$, where the average tax rate τ satisfies

$$\tau \mathcal{B}_t = \mathcal{B}_t - (1-\tau^L)\mathcal{B}_t^{(1-\tau^P)}. \tag{30}$$

After the fiscal rule determines the government debt, and taxes are collected, government expenditure G_t adjusts such that the government budget constraint is fulfilled in every period: $G_t = T_t + B_{t+1} - B_t \frac{R_t^b}{\pi_t}$. As a simplification, it is assumed that G_t does not provide any utility to households. This implies that in steady state, in which government expenditure is calibrated to be strictly positive, a fraction of physical production is wasted.

3.5 Market clearing and equilibrium

The labor market clears at the competitive wage in (22). The market for liquid assets clears when liquid asset demand, which is given by the households' optimal decisions, $L_t^d = \mathbb{E}[\lambda b_{a,t}^* + (1-\lambda) b_{n,t}^*]$, equals the supply of liquidity $L_{t+1} = B_{t+1} + q_t^\Pi$ (as L_t^d is the aggregate over positive and *negative* private liquid asset holdings, the supply of liquid assets is bigger than L_{t+1} in gross terms). Similarly, the price of capital q_t, which is determined by (26), clears the capital market when $K_{t+1} = K_t^d = \mathbb{E}[\lambda k_t^* + (1-\lambda) k_t]$ holds (households that do not adjust capital demand the same amount as last period, k_t). By Walras' law, whenever labor, bonds, and capital markets clear, the goods market also clears.

A *recursive equilibrium* is a set of policy functions $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$, value functions $\{V_t^a, V_t^n\}$, prices $\mathcal{P}_t = \{w_t, w_t^F, \Pi_t^F, \Pi_t^U, r_t, q_t, q_t^\Pi, \pi_t, \pi_t^W, R_t^B, R_t^L, \tau_t, \tau^L\}$, stochastic state A_t and shocks $\Omega_t = \{\epsilon_t, \epsilon_t^l\}$, aggregate capital and labor supply $\{K_t, N_t\}$, distributions Θ_t over individual asset holdings and productivity, and a perceived law of motion Γ, such that

1. Given the functional $\mathbb{E}_t V_{t+1}$ and $\mathcal{P}_t$, the policy functions $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$ solve the households' planning problem, and given the policy functions, $\mathcal{P}_t$, and $\{V_t^a, V_t^n\}$ solve the Bellman equations (10).

2. The labor, the final goods, the bond, the capital and the intermediate good markets clear, and interest rates on bonds are set according to the central bank's Taylor rule.

3. The actual and the perceived law of motion Γ coincide, i.e., $\Theta' = \Gamma(\Theta, \Omega')$.

To solve the model, I use a first-order perturbation around the stationary equilibrium, building on the methods developed by Bayer and Luetticke (2020) and Bayer et al. (2026). Appendix D describes the numerical implementation.

3.6 Definitions and parameter choice

3.6.1 Classification into liquid and illiquid assets

For the classification of assets in the data into the liquid and illiquid categories, I largely follow Kaplan et al. (2014): Illiquid assets, which are assumed to be productive in the model, consist of positive wealth in housing[17], other real estate, pensions and life insurance assets, certificates of deposit, and savings bonds. To compute the net illiquid asset position in the data, illiquid debt is subtracted, namely housing debt on owner-occupied real estate, and other real estate debt. I abstract from car wealth in the analysis[18].

Conversely, liquid assets comprise the sum of checking, savings and call/money market accounts, as well as holdings in mutual funds, equity and other managed assets, and bonds other than savings bonds. For cash holdings, I use the estimate by Kaplan et al. (2014). To arrive at net liquid wealth, I subtract credit card debt. As data source, I use the extension of the Survey of Consumer Finances (SCF), SCF+, by Kuhn et al. (2020), which yields 20 years of cross-sectional data between 1950 and 2016. I restrict the household head to be of working age, i.e., between 22 and 65 years of age.

3.6.2 Parameter choice

The portfolio adjustment probability λ is calibrated at 6.5% so that the mean liquidity in households' portfolios roughly matches the data (see Table 1). This adjustment

[17]This is in accordance with the definition in NIPA, where "the ownership of the house [...] is treated as a productive business enterprise" (U.S. Bureau of Economic Analysis, 2019).

[18]Consumer durables like cars represent a significant share of poorer households' portfolios (e.g., Guiso and Sodini (2013)); however, they are rather evenly distributed across the wealth distribution, so that leaving them out should not bias the results systematically.

Table 1: Calibrations

Targets	Calibration	Data	Source
Mean illiquid assets (K/Y)	11.01	11.44	NIPA
Mean gvmt bonds (B/Y)	1.13	1.66 (1.1)	FRED
Government share (G/Y)	0.18	0.21	FRED
Top 10% wealth share	0.61	0.66	WID
Mean portfolio liquidity	0.25	0.25	SCF+
Fraction without capital	0.13	0.22	SCF+
Fraction borrowers	0.14	0.115	SCF+

Notes: In general, data values denote long-run averages from 1950 to 2016. When subtracting federal debt held outside the U.S. from total federal debt held by the public (data available since 1970), the debt-to-quarterly-GDP ratio of 1.1 is close to the model-implied ratio. The wealth share of the top 10% of the wealth distribution is available from the World Inequality Database since 1962. Portfolio liquidity is defined as the ratio of net liquid wealth to total net wealth. To compute it in the data, I delete all observations of households with positive liquid wealth, but non-positive total wealth (0.7% of total observations). Borrowers are defined as households holding a negative net position of liquid wealth.

probability implies an average waiting time of almost four years until capital holdings can be adjusted. This is also consistent with the interpretation of capital holdings as investments in projects that include R&D, in the following sense: as noted by Li and Hall (2020), the average gestation lag is two years, and the yearly depreciation of R&D in the late 1990s and early 2000s is between 20% and 60% in most sectors[19]. Assuming an initial R&D phase of two years on average, in which intangible capital is produced (while physical capital is pledged as collateral), followed by the phase in which goods are produced using the physical capital and the depreciating intangible capital, I arrive at an average holding time of physical capital of four years. In line with the interpretation of the TFP news shock as an anticipated spillover from intangible capital, I likewise set the persistence $\rho_A = 1.0 - 2 \cdot 6.5\%$, i.e., log-TFP depreciates at a quarterly rate of 13%. The steady state capital share in production is set as in Bayer et al. (2024), $\overline{\alpha} = 0.32$. For the persistence of the shock to the capital share, $\rho_\alpha = 0.9552$, I use the mean probability for firms of losing a low labor-share status within 5 years, as estimated by Kehrig and Vincent (2021).

The size of both news shocks is two times the standard deviation of the respective surprise shocks (see Table 2). For TFP, this is the estimated value from Bayer et al.

[19]Fittingly, Adam and Weber (2023) estimate from product data in the UK the median quarterly turnover rate of consumer products as 13.7%.

Table 2: Estimated parameters (selected)

Parameter	Description	Value
ϕ	Capital adj. costs	0.218
κ	Price stickiness	0.105
μ^Y	Target markup final goods	1.08
κ_w	Wage stickiness	0.133
μ^W	Target markdown wages	1.1
ρ_R	R^B autocorr.	0.803
θ_π	Taylor: inflation	2.614
θ_Y	Taylor: output gap	0.078
γ_B	Fiscal: smoothing	0.157
γ_π	Fiscal: inflation	8.57
γ_Y	Fiscal: output gap	5.73
σ_A	TFP std. dev.	0.00608
σ_α	capital share std dev.	0.005

(2024). For the capital share, I calibrate the size of the news shock to fit the increase in the capital share from the mid 1990s to 2000. To get an estimate of the capital share, I use the NIPA table 1.12 (National Income by Type of Income) and attribute the components to either profit income ($(1-mc)Y$ in the model), wage income (wN in the model), or capital income (rK in the model). Importantly, corporate profits do not enter into capital income (in the model, profit income and capital income are different), while proprietors' income counts towards capital income. While the concrete estimates differ, this exercise is close in spirit to Karabarbounis and Neiman (2019). I find that, between 1995 and 2000, the capital share increased by about 1 percentage point.

The degree of profit smoothing is calibrated to match the standard deviation of quarterly dividend growth of the S&P 500 at $\xi^\Pi = 0.05$[20]. The fractions ω^Π and ι^Π are calibrated to yield a share of liquid assets held in stocks of 31%[21] and a quarterly stock price-dividend ratio of 144[22], which implies $\omega^\Pi = 4.7\%$ and $\iota^\Pi = 0.074\%$. I set $\bar{\eta} = 13.5$

[20]The standard deviation is calculated from simulating the model subject to random innovations in capital share-news shocks, and (surprise) markup and TFP shocks; see Section 5.

[21]From estimations by Saez and Zucman (2016), when defining bonds as fixed income assets plus net deposits and currency, and stocks as equities (other than S corporations), I get a stock share of 45% in 1995. From the SCF wave of 1995 (see e.g., Guiso and Sodini (2013)), when defining bonds as cash and fixed income, and stocks as directly held equity, I compute a stock share of 30%.

[22]This is the mean of the S&P 500 stock price divided by dividends amassed over the quarter, from 1948 to 2016. Its inverse, the dividend yield, implies an average return on stocks *without* capital gains of 2.9% annualized. Net of stock depreciation, the return becomes 2.5% p.a., as for all liquid assets in the model economy.

and $\bar{\zeta} = 11$, which implies price and wage markups of 8% and 10%, respectively. The real liquid rate is chosen to be 2.5% p.a., while the borrowing penalty $\bar{R}$ is set to 7.5% p.a., which leads to the share of borrowers being just a bit higher than in the data (Table 1). The steady state capital rent is $\bar{r} = 3.7\%$ p.a., implying a steady state illiquidity premium of 1.2% p.a. As an estimate for the capital rent, I take the series by Gomme et al. (2011) (including housing, without capital gains, after-tax), which has an average yearly return of 5.6% from 1950 to 2016. Since the model abstracts from long-run technological growth, 2% yearly growth should be subtracted from the counterpart of the illiquid rate in the data. The model liquid asset is composed of both government bonds and riskier equity. Computing real (pre-tax) returns on the S&P stock index, 10-year treasury bonds (data source: Robert Shiller) and 3-month treasury bills, I compute average yearly returns of 8.3%, 2.5%, and 0.7%, respectively, over the period from 1950 to 2016. The liquid rate in the model should be considered as a weighted average of these rates[23].

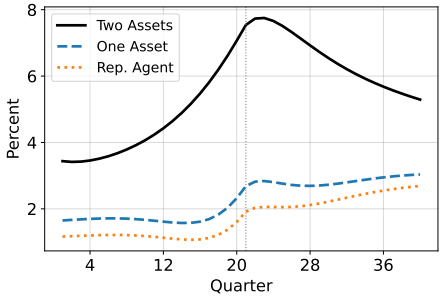
Tax progressivity $\tau^P = 0.18$ is taken from Heathcote et al. (2017), while the tax level $\tau^L = 0.1$ is set to achieve a government share of roughly 18%. With respect to the parameters that Bayer et al. (2024) estimated, I choose those estimates where inequality data were included in the estimation (the HANK* specification). Importantly, I deviate with respect to the fiscal rule, where I estimate γ_π and γ_Y so that the ratio of the magnitude of the profits- and the magnitude of the bonds-response in the anticipation phase of the news shock matches the respective ratio in the late 1990s[24]. Table 2 lists the chosen values for a selection of parameters in the model.

[23]The introduction of aggregate risk, which would allow differentiating among the classes of liquid assets by their model-implied riskiness, would be an advantage for this part of the calibration. For stock holdings, one should account for the capital gains tax rate of 15-25% over the sample for wealthy households, and discount dividends by 2% long-run growth. Additionally, the financial intermediation wedge of 1.5-2% as calculated by Philippon (2015) reduces the effective rate of financial assets for households.

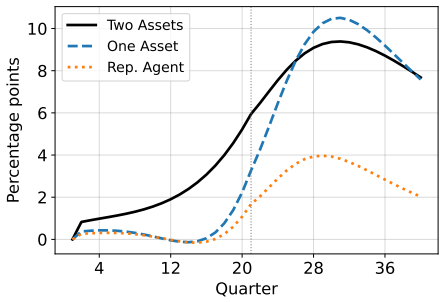
[24]I define the magnitude of the impulse response as the distance between the maximum and the minimum of the percent deviations in the anticipation phase. I constrain both γ_π and γ_Y to the interval $[-10.0, -0.01]$, and search for a global minimum using a simulated annealing algorithm. The estimated bond supply is much more elastic, i.e., the government stabilizes inflation and the output gap more aggressively, than what was estimated by Bayer et al. (2024) for the whole period since 1960. The reason is that in the late 1990s, the U.S. government strongly reduced its debt.

Figure 3: Response of stocks across model classes

a. Stock price $[q^{\Pi}]$



b. Share of stocks within liquid asset $[q^{\Pi}/L]$



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21).

4 A news-induced stock price cycle

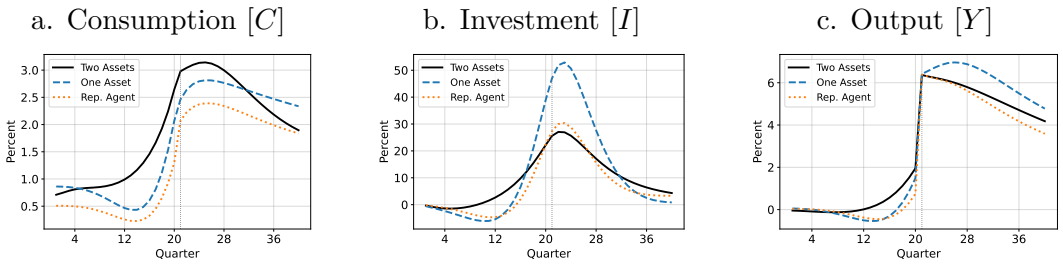
I consider the following experiment: with an anticipation horizon of 5 years ($\ell = 20$)[25], households become aware that the capital share will increase (by two times its standard deviation). As outlined in the introduction, one can interpret the capital share increase as a temporary change in the production process due to, e.g., more firms employing IT capital. In Section 4.2, I show that I obtain almost the same impulse responses if the news is instead about a temporary increase in TFP. The reason is that for both news shocks, the expectation of a higher future return on holding capital is identical, which is the decisive impulse to cause the investment-driven boom. The higher expected lifetime income that induces households to increase their consumption in the anticipation phase is mainly produced by the higher capital stock, which is accumulated in both scenarios when households rebalance their portfolios towards the productive asset.

Figures 3 and 4 present the response of the stock price and business cycle variables across three model variants: *Two Assets* denotes the baseline model with heterogeneous agents and portfolio choice between liquid and illiquid assets. *One Asset* retains the market incompleteness, but takes away the portfolio choice: every household holds a representative portfolio, which is determined by the bond supply rule and the ex-ante illiquidity premium being fixed at a steady state level of zero[26]. This implies that capital becomes liquid in this setting. *Rep. Agent* additionally takes away market incompleteness,

[25]I choose an anticipation horizon of five years to be close to the dot-com boom example: Karnizova (2012) estimates increased "productivity prospects" around 1995, while in 2000, the NASDAQ peaks.

[26]The ex-ante illiquidity premium is defined up to first order as the difference between the expected return on capital and the expected return on liquid assets, $\frac{\mathbb{E}_t(q_{t+1}+r_{t+1})}{q_t} - \frac{R_t^b}{\mathbb{E}_t\,\pi_{t+1}}$.

Figure 4: Response of the business cycle across model classes



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21).

and is thus a model of the RANK variety[27].

Only the HANK model with portfolio choice exhibits a peak in the stock price around the time of the capital share increase (quarter 21), and generates the uniformly accelerating stock price growth that is typical for stock price booms. It is clear that the decisive difference for whether the news drives the business cycle is the portfolio choice. In the full HANK model, richer households start shifting their portfolio towards the illiquid capital after around 2.5 years. This crowds out government bonds (which increases the share of stocks within liquid assets) and thus government expenditure. The higher demand for goods increases wages (since prices are sticky) and lowers the negative labor gap (since wages are sticky), so that households increase their labor supply. Aggregate consumption rises on impact as households expect to have a higher lifetime income, and increases gradually with higher incomes. This gradual consumption increase (by most households) supports a higher real interest rate in equilibrium.

Figure 5 shows the response of the (ex-post) returns to the two asset classes, liquid and illiquid assets, across the model varieties. It is clear that without a time-varying illiquidity premium, the expected returns are the same across asset classes (the liquid asset return jumps up at the onset of the news, as the stock appreciates unexpectedly). In contrast, with illiquid capital, the illiquidity premium declines during the anticipation period (the return on liquid assets increases) and rises after the stock price peak (the return on liquid assets falls). Panel c) shows the effect of the illiquidity premium on the demand for capital at the extensive margin. This margin is only active in the bottom half of the wealth distribution: in steady state, 26% of these households hold no capital, while practically all households in the upper half do. While richer households increase their

[27]The households' time preference β is calibrated in the RANK and the One Asset varieties such that the real rate on the asset in steady state equals that of the baseline Two Asset model. This implies that the steady state stock price-dividend ratio is also equal across all three varieties.

capital holdings during the anticipation phase (intensive margin), the lower illiquidity premium deters poorer households from holding capital: the share of households without capital in the bottom half rises by 2 percentage points within the first three years after the news. As the capital share increase approaches, this share declines again. After the boom, when the illiquidity premium rises, it falls below its steady state level, i.e., the demand for capital rises, which increases the capital price.

The increasing real interest rate in the anticipation period does not depress the economy; to the contrary, it stabilizes the income of richer households by increasing their return on liquidity, which enables them to invest in capital, inducing the boom. Households in the top 10% of the wealth distribution own 65% of the capital stock in the economy, and they account for most of the capital accumulation during the anticipation phase. However, if the profit losses of entrepreneurs were higher, or interest income lower, more of the richest households would sell capital to offset their income losses, thereby depressing aggregate investment.

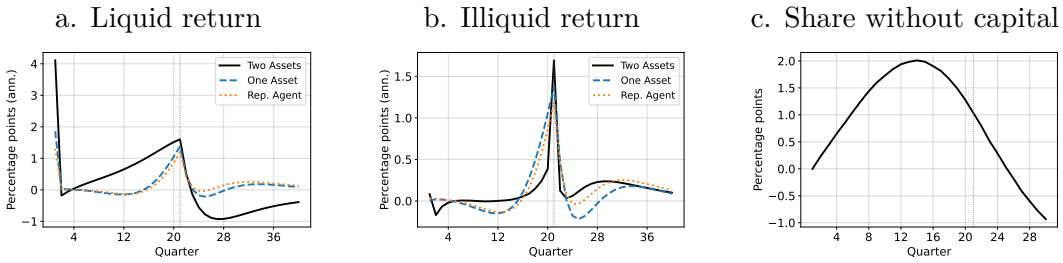
4.1 Comparison to the dot-com boom

Since both the capital share shock and several parameters were calibrated to the 1990s in the U.S., I can make a quantitative comparison of the shock responses to the aggregate observations from 1995 to 2000[28]. In terms of real business cycle variables, the model replicates the 6% rise in output, while it overstates the 15% increase in investment and only accounts for one third to one half of the observed increase in consumption. As noted above, I calibrate the fiscal rule so that the model responses match the ratio of the decline in U.S. government debt to the decline in corporate profits during the late 1990s. In absolute size, the model explains between 75% and 85% of the observed declines in government bonds and profits (notably, federal debt held by the public declined by 20% during that time).

The shortcoming with respect to aggregate consumption may be due to the fixed debt limit in the model, while in reality, financial innovation related to collateral borrowing might have allowed households to consume more. Considering only unsecured borrowing, I find that the model accounts for half of the 30% increase in consumer credit. In the model, the increase in borrowing, mostly by the bottom 50% of households, contributes to

[28]I detrend all time series by a constant growth rate of 2%, following McGrattan and Prescott (2010), and deflate nominal series with the GDP deflator [GDPDEF].

Figure 5: Response of *ex-post* returns and capital holding across model classes



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21). The return on capital includes capital gains. Panel c) shows the bottom 50% of the wealth distribution, defined in the *cross-section* each quarter.

the overall increase in wealth inequality during the anticipation period. From the World Inequality Database, the Gini index of wealth increased by 1.25% in that time span; the model explains a little more than a third of this increase[29]. Finally, with respect to the share of stocks within the liquid asset class, using the estimates by Saez and Zucman (2016), during the dot-com boom this share increased by 20 percentage points. The model accounts for around 25% of this increase[30].

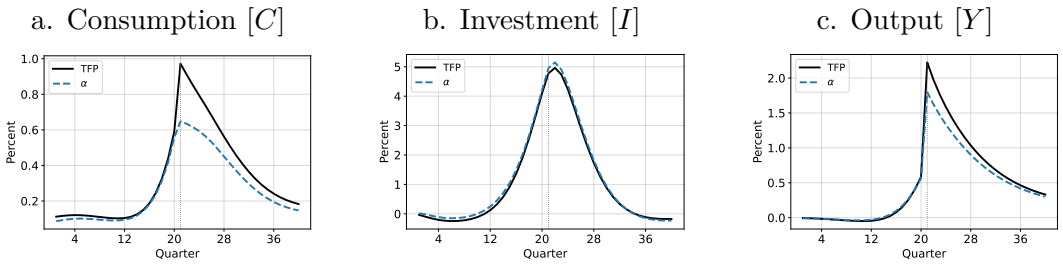
4.2 Alternative news shock

Figure 6 compares the response of the business cycle to news about a temporary TFP increase with the response to the capital share news (I adjust size and persistence of the shocks to make them comparable). The responses are virtually identical in the anticipation phase. This shows that the portfolio rebalancing towards capital, which is incentivized in both cases by the expectation of higher future returns on holding capital, drives the boom also in consumption and output. Differences only occur once the fundamental shock materializes: a higher capital share redistributes from households with a high marginal propensity to consume to those with a low propensity, so that consumption rises by less, while higher TFP implies more income for all households. Therefore, output also rises a little less in the case of the capital share increase. Still, in the long run, the levels of consumption and output converge across the two shock responses. The reason is that, when the direct effect of the transitory shocks subsides, the indirect effect of the higher

[29]This is remarkable, since the model does not feature heterogeneous stock shares; in the data, rich households gain disproportionally from stock price booms, see Kuhn et al. (2020).

[30]A more detailed model of stocks and their difference compared to other liquid assets, namely the different aggregate risk they carry, could help explain this gap. Institutional changes, or agents that learn about the fundamentals over time, receiving observed prices as signals, would be other possible explanations.

Figure 6: Response of the business cycle to an alternative news shock



Notes: Model impulse responses are to news about a temporary TFP increase in 5 years, and to news about a temporary capital share increase in 5 years (both quarter 21). The size of the capital share impulse is scaled to fit the TFP news shock (given by two times σ_A). For comparability, the persistence of the capital share process is adjusted to ρ_A.

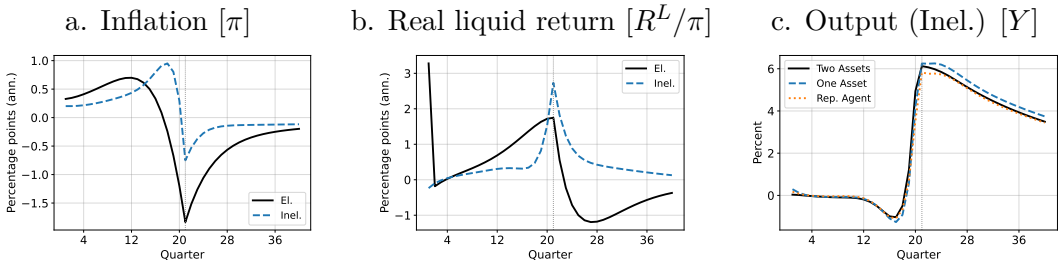
capital stock, built up during the identical anticipation phase, dominates.

In a further clarifying exercise, I also shock the model economy with news about future transitory increases in the markup μ (i.e., market power), and news about future increases in investment-specific technology productivity, which increases the marginal productivity of the transformation from consumption to capital goods. Both variables are prominent candidates in the literature to explain the secular decline (increase) in the labor (capital) share (e.g., in Karabarbounis and Neiman (2014), Greenwald et al. (2019)). I find that both news shocks depress the economy in the anticipation phase. The markup shock implies an expected redistribution from capital to profit income, which disincentivizes the holding of capital, so that investment falls. On the other hand, the investment-specific technology shock increases the capital rent, but it lowers the cost of capital; therefore, households postpone investment until capital becomes cheap. This illustrates how only the anticipation of high rents *and* returns for capital causes an investment-driven business cycle and stock price boom in the model.

4.3 Importance of the fiscal rule

The investment boom is enabled by an elastic bond supply and a government that is willing to temporarily reduce its expenditure. To illustrate this point, I compare the response of inflation and the real liquid return in the baseline model with the impulse responses in an alternative environment (*Inel.*), where the government does not stabilize the output gap, and stabilizes inflation less strongly (Figure 7). With the alternative fiscal rule that allows for a prolonged rise of inflation during the anticipation phase, households do not invest enough to start the business cycle (and stock price) boom. The reason is that inflation depresses asset returns and magnifies the increase in the marginal costs

Figure 7: Responses for different bond supply elasticities



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21).
The changed fiscal rule parameters of the *Inelastic* specification are $\gamma_Y = 0.007, \gamma_\pi = 6.58$.

of firms (affecting the entrepreneurs) and of unions (affecting the workers) late in the anticipation phase. The expectation of being exposed to these income losses discourages the households' capital investment earlier in the cycle. As a result, even in the model with portfolio choice, government expenditure is crowded out too late to drive the boom, and therefore all three model variants exhibit roughly the same output response (as well as consumption response) to the news shock.

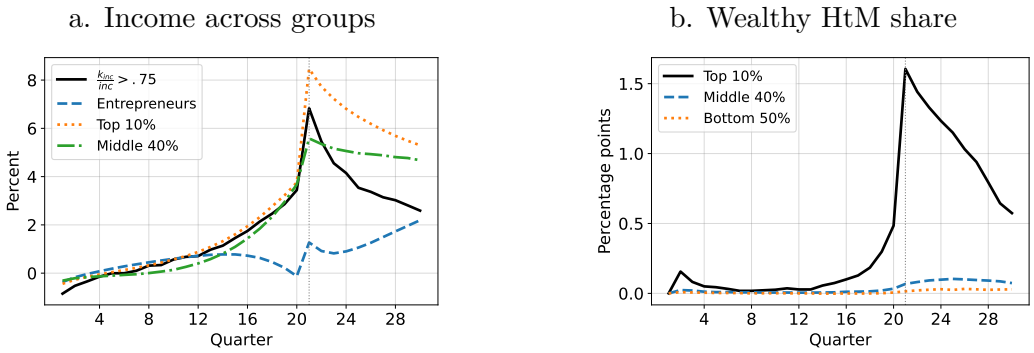
4.4 Wealthy hand-to-mouth households

Following Kaplan et al. (2014), wealthy hand-to-mouth households are households that have non-zero wealth in the illiquid asset ($k_i > 0$), while being at a kink in the budget set: either at zero liquid savings ($b_i = 0$), or at the borrowing limit ($b_i = \underline{B}$). Motivated by my numerical findings, I focus on the case when households hold the illiquid asset, while being at the borrowing constraint. Kaplan et al. propose a stylized 3-period life-cycle model without uncertainty to highlight the conditions under which it is optimal for households to be wealthy hand-to-mouth: Suppose that in the first period, households allocate their initial endowment between the liquid and the illiquid asset. Next period, they receive income and can sell their liquid asset (or borrow) to increase their consumption, but cannot sell the illiquid asset until the third (and last) period, where they consume their income and the return to all asset holdings.

In this setup, households are more likely to be wealthy hand-to-mouth at the end of the second period if:

1. the capital rent and price in the last period are high relative to the borrowing rate,
2. their initial endowment is high, and both capital rent and their income are increasing from the second to the last period.

Figure 8: Response of income and shares of wealthy hand-to-mouth



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21).
Left panel: $\frac{k_{inc}}{inc} > .75$ denotes households whose main source of income ($> 75\%$) is capital rents ($[r]$ in the model). All groups are defined in the *cross-section* each quarter.
Right panel: Wealth groups are defined from their position *at period 0*.

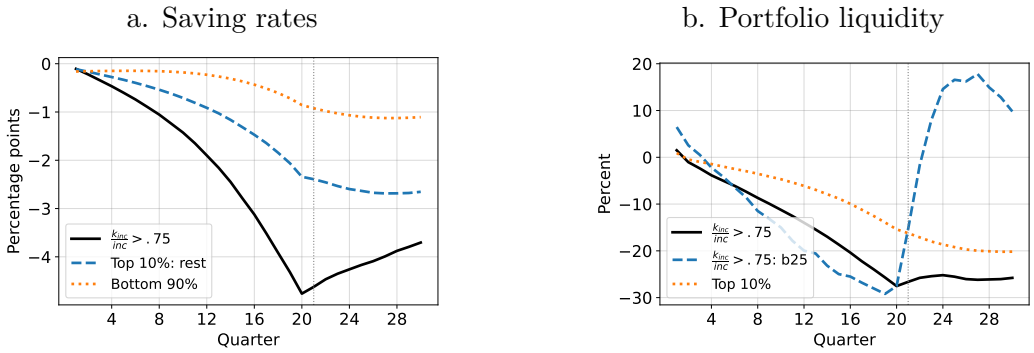
The news shock raises the expected capital rent and prices in the future. As I argued in Section 4.3, extreme profit swings towards the end of the cycle depress investment. Part of the reason is that a big output gap late in the cycle requires monetary policy to hike the nominal rate, so that the real rate spikes in the last quarter before the capital share increase. This makes it more expensive to finance illiquid asset holdings with debt accumulated over the anticipation period, so that more households will refrain from doing so (as discussed above, higher real rates *earlier* in the cycle instead are beneficial for investment).

While the income of the average household in the upper half of the wealth distribution rises during the stock price boom, households whose income is dominated by capital rents gain about as much as the average household in the top 10% (see Figure 8a). Entrepreneurs, who receive the profit income, experience an income rise at the onset of the capital share increase, but gain little in the anticipation period and lose towards its end. Therefore, entrepreneurs are less likely to become wealthy hand-to-mouth households in the anticipation phase[31]. Hence, by virtue of capital rents, holding (a high amount of) the illiquid asset and experiencing income gains reinforce each other, making point 2) more likely to hold.

For these reasons, it is mostly households at the top of the wealth distribution who become wealthy hand-to-mouth households during the anticipation phase (see Figure 8b). In steady state, less than 0.1% of households are wealthy hand-to-mouth (at the borrowing

[31]What is more, entrepreneurs on average hold much larger liquid asset stocks than workers, as they face the largest idiosyncratic risk (becoming a worker).

Figure 9: Response of portfolio choice across groups of households



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21).
Left panel: The saving rate is defined as $1 - c_{it}/\{\text{cash at hand}_{it}\}$, where

$$\text{cash at hand}_{it} = y_{it} + b_{it}R_t^L/\pi_t + k_{it}(r_t + \mathbb{1}_{\{k \text{ adjustable}\}}q_t) - \underline{B}.$$

All groups are fixed in quarter 20, the quarter before the capital share increases: $\frac{k_{inc}}{inc} > .75$ denotes households whose main source of income ($> 75\%$) is capital rents, *Top 10%: rest* the other households in the top 10% of the wealth distribution, and *Bottom 90%* the households below.
Right panel: Portfolio liquidity is with respect to the *chosen* portfolio, i.e., households' wealth position next period. $\frac{k_{inc}}{inc} > .75$ denotes households whose main source of income ($> 75\%$) is capital rents ($[r]$ in the model). $\frac{k_{inc}}{inc} > .75 : b25$ denotes the mean of the *lowest quartile* of the portfolio liquidity distribution for these households. All groups are defined in the *cross-section* each quarter.

limit). 64% of those households are in the top 10% of the wealth distribution. I find that during the stock price boom, the share of wealthy hand-to-mouth households among the wealthiest decile roughly doubles until the end of the anticipation phase, and quadruples when the capital share increases. Hence, by far the largest inflow into the group of wealthy hand-to-mouth households comes from capital-wealthy households, who optimally choose to get at or near the borrowing constraint so that they can hold on to the capital a little longer.

4.5 Marginal traders

How can one tell whether the mechanism highlighted in Section 2 is at work in the full HANK model? The mechanism implies that the households who hold large, illiquid capital positions when the capital share increases become the marginal savers: exposed to high consumption risk, they increase their liquid saving, which depresses the return on liquid assets. To show this, I fix the group of households whose income is dominated by capital rents in the quarter before the capital share increases (quarter 20), a group that grows during the anticipation phase, and follow their saving rate (defined as the fraction

that is saved of all funds available to the household in a given period) over the whole cycle. Figure 9a compares it to the saving rates of the rest of the top 10% and of the bottom 90% of the wealth distribution in that quarter. The "rentiers" reduce their saving rate most strongly during the anticipation period. After the capital share increase, they are the only group of households whose saving rate is trending upwards, which indicates that these households drive down the return on liquid assets. Note that, since the aggregate supply of liquid assets is down, also a saving rate below its steady state level can depress the return on liquid assets in equilibrium.

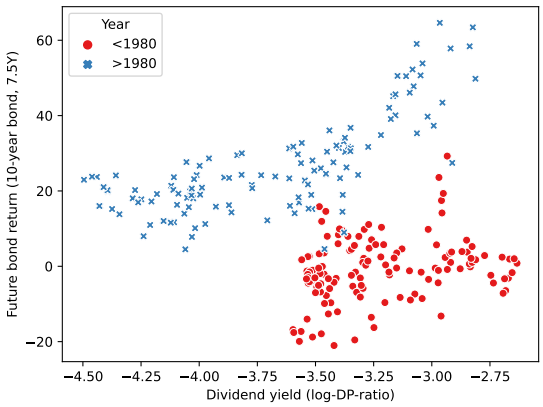
Figure 9b shows the portfolio liquidity of households in the richest decile in the cross section. Among the rich households, it is the households whose income is dominated by capital income who decrease their portfolio liquidity early on. During the anticipation phase, the distribution of portfolio choices of households with dominant capital income widens. One of the reasons is a *composition effect*: households with less capital wealth enter the group by virtue of higher capital rents during the business cycle boom. This alone drives up the portfolio liquidity of households in this group compared to the steady state[32]. Therefore, I also show the mean response of the lowest quartile in the portfolio liquidity distribution of these households. The marginal traders will be in this region of the distribution during the anticipation phase. I find that households with high capital income in that region of the distribution lower their portfolio liquidity during the anticipation phase. After the boom, the "rentiers" in that region of the distribution increase their liquid saving—their portfolio liquidity rises—as they are exposed to high consumption risk at that point. This depresses the real rate on liquid assets in equilibrium.

5 Asset returns, heterogeneous portfolio choices, and the stock market

In this section, I provide empirical evidence for the relation between the returns on liquid and illiquid assets and stocks, and the relation between portfolio choices of households and stocks, using micro-level data. Then, I simulate the model in order to assess the quantitative success of the model in explaining stock price fluctuations. Additionally,

[32]In the data, this composition effect rather goes in the opposite direction: since empirically, capital rents increase less in stock price booms than real bond rates, there is some evidence that the overall share of households with dominant capital income decreases in stock price booms. However, this does not drive the overall reduction in portfolio liquidity: see Section 5.

Figure 10: Liquid return news and dividend yield



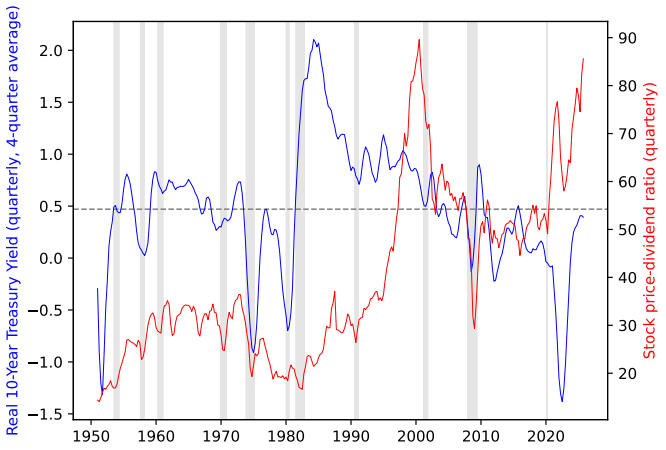
Notes: Data by Robert Shiller (S&P and 10-year treasury bond). All returns are ex-post (realized) quarterly observations from 1955.Q4 to 2016.Q4. The future real 10-year treasury yield is a weighted sum over the next 7.5 years, with discount factor $\rho = 0.96$ computed as in Campbell and Shiller (1988). Ex-post real returns are computed using realized inflation.

I use the Campbell-Shiller decomposition of the model stock price to highlight the effects of different assumptions about the cyclicality of dividends and the accuracy of the news for the explanatory power of the mechanism.

The theory implies that news about future returns on liquid assets, like government bonds, causes today's dividend yield of the stock market to rise. The asset pricing literature (Cochrane, 2005) finds that this relation is instead negative, and that therefore the discount rate channel exclusively runs through excess stock returns (the "risk premium"). Figure 10 shows visually that this finding flips when accounting for a secular break in 1980: real bond returns fell in the 1970s due to high inflation, implying mostly negative liquid return news before 1980. At the same time, the dividend yield was higher in the earlier part of the sample. This stark secular pattern causes the relation over the whole sample to be negative. However, splitting the sample in 1980 and running a Campbell and Shiller (1988)-decomposition on the two time periods separately, I find that future bond returns correlate positively with the dividend yield, and explain part of its variance, especially in the period since 1980.[33] Plotting the real 10-year treasury yield, averaged over four quarters, together with the stock price-dividend ratio over time gives an impression of the relevance of this correlation as evidence for my theory (Figure 11). It shows that the larger swings in stock prices in the last decades, namely the downturns in the 1970s

[33]Concretely, I find that the share of the variance of the S&P 500 dividend yield explained by future bond returns amounts to 8% in the pre-1980 sample (compared to 81% variance share explained by excess return news), while future bond returns explain 21% of the dividend yield since 1980 (compared to 29% for excess return news).

Figure 11: Real 10-year treasury yield and the stock market



Notes: Stock market data from S&P 500 (Robert Shiller), recession years (grey areas) by NBER. The real 10-year treasury bond yield is the nominal yield minus the inflation rate over the preceding twelve months, shown as a four-quarter moving average. The dotted line marks the average quarterly real 10-year treasury bond yield over the sample (0.47 pp).

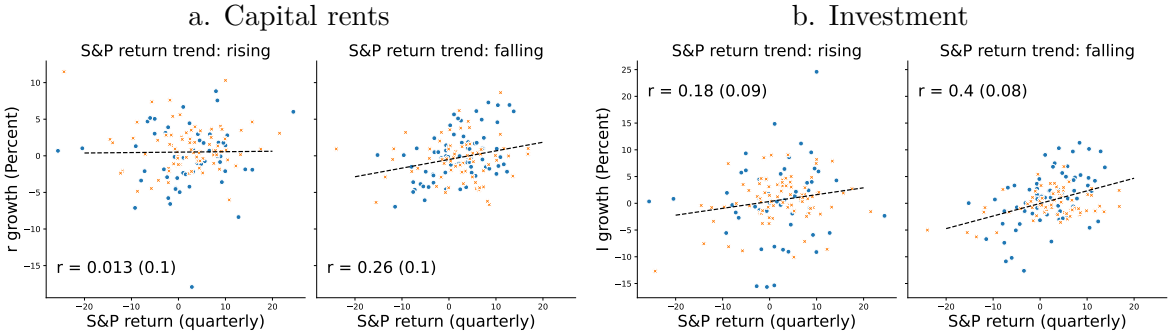
and 2000s, and the booms in the 1980s and 1990s, occur in times of lower-than-average and higher-than-average real interest rates, respectively.

Next, I take the capital return series by Gomme et al. (2011) as a proxy for returns on illiquid assets (no capital gains, after-tax), and examine whether the change in capital returns is related to stock returns, as in the theory. Specifically, the proposed mechanism hinges on capital-wealthy households driving down the return on liquid assets, and thus also stock returns, when capital returns fall. Figure 12a shows the correlations. During the boom phase, there is no correlation, but when stock prices are falling, there is a weak correlation. For investment growth, the correlations are more strongly positive. In a regression exercise (see Appendix C.1), I check that the positive correlations are unaffected by the inclusion of dividends and other business cycle variables. In sum, the data are consistent with a theory of investment-driven stock price booms, where a fall in capital rents depresses stock returns after the boom.

5.1 Evidence from survey data

Turning to heterogeneous portfolio choice, which is a crucial part of the proposed theory, I use the SCF+ by Kuhn et al. (2020), extended with the 2022 and 2025 waves of the Survey of Consumer Finances, to isolate the group of households for whom capital income (excluding capital gains) is the main share (at least 75% in the baseline) of their

Figure 12: Capital rents and investment over the stock price cycle



Notes: Data by Robert Shiller (S&P 500) and Gomme et al. (2011) (capital rents). Quarterly observations (1948.Q2–2016.Q4). S&P return trend computed using the HP filter ($\lambda = 1600$). Blue dots: before 1980. Orange crosses: after 1979. Newey-West standard errors (1 lag) in parentheses.

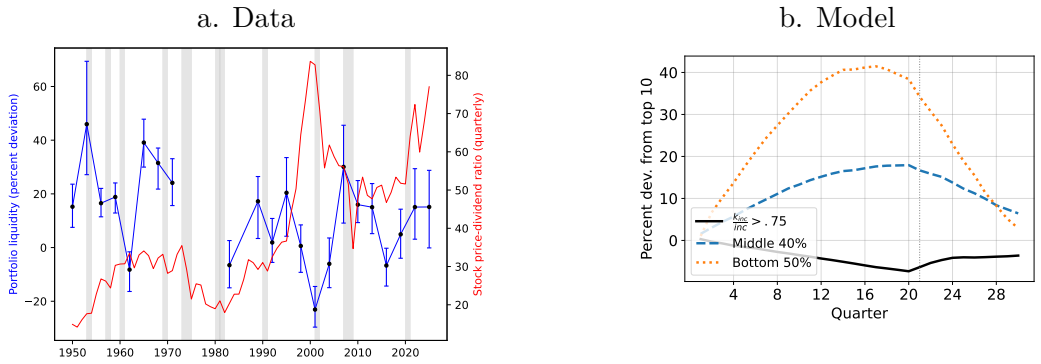
overall income, and who do not actively manage a business[34]. I refer to these households as "rentiers": they live off the income of illiquid assets that they hold passively. I discuss at the end of this section why I leave out the owner-managers of private businesses. On average over all sampled years, 1.4% of households are in that category (2.3% in the model). In the model, the portfolio choice of such households is decisive in affecting the illiquidity premium, and thus stock prices, over the cycle. In order to abstract from secular trends in the portfolio liquidity of the different wealth groups, I take the relative portfolio liquidity of the households with high capital incomes compared to the portfolio liquidity of the top 10% of the wealth distribution as the main measure of comparison between model and data[35].

While in the model, households with high capital incomes are all in the top decile of the wealth distribution, in the data, only 30% are in that wealth group, while 43% have wealth that lies between the median and the top 10% of the wealth distribution. The likely reason for this discrepancy is that the model abstracts from negative illiquid assets: mortgage debt in particular systematically lowers the net worth of households with high capital income in the data. Due to this overlap of the "rentiers" with lower wealth groups, I also compute the relative portfolio liquidity of the bottom 50% and middle 40% relative to the top 10%. This allows me to see if movements in the relative portfolio liquidity of

[34]In the older waves of the Survey of Consumer Finances before 1983, capital income is only available as a measure that lumps together income from illiquid and liquid investments (like dividend income), while only the former counts as capital income in the model. Therefore, I treat the time periods before and after 1983 separately. Whether a household actively manages a business is also known from 1983 on. See Appendix C.2.

[35]I show the time series of the portfolio liquidities of the different groups, as well as other characteristics of their portfolio choices over time, in Appendix C.2.

Figure 13: Relative portfolio liquidity in model and data



Notes: Survey evidence from SCF+ (Kuhn et al., 2020) and the SCF waves of 2022 and 2025, stock market data from S&P 500 (Robert Shiller), recession years (grey areas) by NBER. Portfolio liquidity is defined as the ratio of liquid assets to total wealth.
Left panel: Left axis shows the relative deviation of portfolio liquidity of households whose main share of income (>75%) is capital income and who do not actively manage a business, from the portfolio liquidity of the top 10% of the wealth distribution. Whiskers are 68%-confidence intervals.
Right panel: Model responses of relative portfolio liquidity deviations (with respect to top 10%) across groups in the cross section. Responses are net of steady state deviation.

the households with high capital incomes are spuriously driven by movements across the wealth groups.

The left panel of Figure 13 shows the relative portfolio liquidity of "rentiers" over time, and in comparison to the stock price-dividend ratio of the S&P 500. The right panel shows the model-implied prediction of the relative portfolio liquidities following a news shock. The model predicts that in response to the news, households with high capital income reduce their portfolio liquidity relative to the top 10%, and increase it again in the years after the boom. Households in the bottom 90% of the wealth distribution reduce their portfolio liquidity by less than the top 10%, so that their relative portfolio liquidity rises in the anticipation phase and falls after the boom.

To put this prediction to the test, I conduct the following exercise: let $\{y_i\}_i$ denote the sequence of consecutively sampled years in each of two subsamples of the survey: years between 1950 and 1971, and years between 1983 and 2025. For each sequence of the relative portfolio liquidities of households in group g, computed from the survey data, denoted by $\{\text{pflq}^g(y_i)\}_i$, I compute the difference between subsequent years: $\Delta_i \text{pflq}^g = \text{pflq}^g(y_i) - \text{pflq}^g(y_{i-1})$. I also collect the stock price-dividend ratios for the years in which survey data are available, and compute the same differenced sequence, $\Delta_i \frac{q^\Pi}{\Pi^F} = \frac{q^\Pi}{\Pi^F}(y_i) - \frac{q^\Pi}{\Pi^F}(y_{i-1})$. Then, I combine the differenced variables of both sets of years into one pooled sample. Column (I) in Table 3 shows the results

Table 3: Regression of price-dividend growth on relative portfolio liquidities

Variables	(I)	(II)	(III)	(IV)	(V)
high cap. inc.	-0.29 (0.18)	-0.29 (0.22)	-0.56** (0.21)	-0.34* (0.19)	-0.58** (0.19)
middle 40%	0.44** (0.18)	-0.07 (0.09)	0.52** (0.18)	0.63** (0.21)	0.67** (0.20)
bottom 50%	-0.77** (0.21)	-	-0.82** (0.21)	-0.81** (0.19)	-0.85** (0.20)
rel. stock share	-	-	0.52** (0.14)	-	0.49** (0.11)
in top 10% share	-	-	-	0.34 (0.19)	0.28* (0.15)
Adj. R-squared	0.29	-0.03	0.47	0.33	0.50

Notes: The baseline regression equation is $\Delta_i \frac{q^\Pi}{\Pi^F} = \alpha + \sum_g \beta_g \Delta_i \text{pflq}^g + \epsilon_i$, $i = 1, .., 20$. I divide all variables by their standard deviation. Specifications (III) and (V) include the change in the ratio of the stock share of high-capital-income households to the stock share of households in the top 10% as a regressor. Specifications (IV) and (V) include the change in the share of high-capital-income households in the top 10% as a regressor. Newey-West (one lag) standard errors in parentheses. Asterisks indicate that the t-statistic of the coefficient is above the 5% (**) or 10% (*) level.

of regressing $\Delta_i \frac{q^\Pi}{\Pi^F}$ on the change in relative portfolio liquidity $\Delta_i \text{pflq}^g$ of the groups $g \in \{\text{high capital income}, \text{middle } 40\%, \text{bottom } 50\%\}$. As predicted by the model, the relative portfolio liquidity of households with high capital income comoves negatively with stock price-dividend growth, with a coefficient of -0.29 (standardized) when controlling for the portfolios of the other two wealth groups, although the estimate is not statistically significant at the 10% level in this baseline specification. Notably, the relative portfolio liquidity of the poor half of the wealth distribution also correlates negatively with the stock market. Appendix C.2 shows that this is due to secular patterns in the ratio of the portfolio liquidity of the bottom 50% and the top 10% over the sample that correlate with secular trends in the stock price-dividend ratio. I find that the portfolio liquidity of the "rentiers" and the bottom 50% explain mostly different parts of the variation, as leaving the latter out of the regression yields largely the same result for the "rentiers".

One potential issue with the interpretation of the results is that they could arise mechanically, through a composition effect with respect to stock shares: On average over the sampled years, households in the top 10% of the wealth distribution hold 12.9% of their total wealth in stocks, while households whose income is dominated by capital income hold 10% of their wealth in stocks[36]. Since stocks are liquid, the higher valuation of stock wealth during stock price booms mechanically increases the liquid wealth and, ceteris paribus, also the portfolio liquidity of the top 10% relative to the households with high capital incomes. To check if this mechanism drives the results, I add the relative

[36]The share of wealth that the top 50% of the wealth distribution holds in stocks decreases markedly from the first to the second half of the sample, see Appendix C.2.

stock share of the "rentiers" compared to the top 10% as an additional regressor, where the stock share is defined as the ratio of the wealth held in equity and other managed assets to the total wealth of the household. Columns (III) and (V) in Table 3 show the results. When controlling for the stock share, the evidence for a negative relation between the relative portfolio liquidity of high-capital-income households and the stock market becomes stronger. The reason is that the relative stock share of the "rentiers" varies strongly from one survey wave to the next, as the group is small, and that waves in which they hold more stocks are waves in which their portfolio is more liquid. This variation is only weakly related to the stock market, and blurs the negative relation between the relative portfolio liquidity and the stock market in the baseline specification.

To interpret the results as evidence for portfolio choice, one should also account for another composition effect: As shown above, stock price booms coincide with higher returns on liquid assets and business cycle booms. Hence, the overall income of households rises on average in stock price booms. If at the same time, capital rents do not rise (as much), the share of households whose income mainly comes from capital income falls. As a consequence, those households that *remain* above the threshold (>75% of income is capital income) have higher illiquid wealth, and thus a lower portfolio liquidity. The negative correlation of the portfolio liquidity of those households with the stock market would then be a mere restatement of the relation between the stock price cycle and factor incomes[37]. In columns (IV) and (V), I consider this possibility by including the change in the share of households with dominant capital income within the top 10% as an additional regressor. I find no evidence that the share of "rentiers" among the wealthiest households is countercyclical—if anything, it rises with the stock market—and the negative relation between relative portfolio liquidity and the stock market remains. To summarize, the evidence from the Survey of Consumer Finances is in line with the predicted fall in the liquidity of the portfolios of households with high capital incomes during stock price booms, and the relation is statistically significant once the stock share is controlled for.

5.1.1 Who are the "rentiers"?

The survey data can be used to investigate the main characteristics of the "rentiers". For this, I take averages over all sampled years for the "rentiers" and the rest. I find that 49% of the "rentiers" report no wage income, compared to 20% of the rest. Only 9% of

[37]Note that, in the household survey, capital gains from equity do not count as income. I use the same accounting in the model. Therefore, stock price booms do not mechanically raise liquid incomes.

the "rentiers" report positive income from self-employment or a business, while among the rest of households, 20% report such income, and 25% are professionals or managers, compared to 31% of the other households. By construction, none of the "rentiers" actively manages a business from 1983 on, and business wealth makes up only 3% of their wealth, against 6% on average for the rest of the households. With their high capital income, 19% of these households are in the top 10% of the income distribution.

"Rentiers" are thus passive holders of illiquid capital: they live off rents, royalties, trust income, and the income of businesses that others run. I leave owner-managers out of the group for two reasons. The main reason is that the capital of their firm is not illiquid for them: through their labor effort and their accounting decisions, owner-managers can actively shift resources between the firm and the household. Most of the profit of pass-through firms is a return to the work of their owners (Smith et al., 2019), and Holm et al. (2026), for example, find that the owners of private firms in Norway adjusted their spending to news about a future dividend tax with resources that they moved between their firm and their household. The holder of a rented building, of a trust, or of a passive interest in a partnership cannot draw on that wealth at will, and is exposed to the consumption risk that drives the mechanism of Section 2. The second, lesser reason is to keep the group comparable across the waves of the survey: since 2004, the survey counts the income of owner-managers from partnerships and S corporations as capital income, and as business income before (see Appendix C.2).

5.2 Simulation

In order to evaluate the ability of the quantitative model to explain unconditional moments of the stock market and its correlation with the business cycle, I simulate the model. For the baseline, I pick three shocks: surprise TFP shocks ϵ^A, a surprise shock to the target price markup μ^Y, and the capital share news shock at the 5-year horizon, $\epsilon^{\alpha,20}$. I set the standard deviation of the price markup shock to 0.01645, taken from the estimation by Bayer et al. (2024). In one simulation variant, I implement a *noise shock* instead of a news shock. In practice, this is achieved by adding a surprise capital share shock ϵ^α to the system in every period where the capital share change was expected to take place. The surprise capital share shock exactly offsets the effect of the capital share news shock (Chahrour and Jurado, 2018). In other simulation variants, I leave out any anticipatory shocks. I assume all shocks to be normally distributed around zero.

Table 4: Unconditional moments in data and simulated model

Variables	Data	(I)	(II)	(III)	(IV)	(V)
$mean(P/D)$	152*	151	149	147	147	150
$\sigma(P/D)$	63	43	33	28	27	39
$\rho(P/D)$	0.98	0.99	0.989	0.994	0.996	0.972
$\rho(\Delta P/D)$	0.99	0.14	0.03	0.38	0.37	-0.04
$\sigma(\Delta D)$	1.75%*	1.74%	1.31%	1.81%	1.46%	1.46%
$\rho(I/Y, P/D)$	15.2%	45%	13%	-1%	-21%	25%
$\rho(\Delta I/Y, \Delta P/D)$	17.5%	38%	31%	12%	-11%	65%
$\rho(\Delta C/Y, \Delta P/D)$	15.4%	9%	-31%	11%	-56%	70%
$\rho(R^b/\pi, R^{stocks})$	0.13-0.19	0.25	0.24	0.11	0.00	0.24
$\sigma(R^{stocks})$	7.28%	3.81%	3.35%	1.75%	1.52%	6.02%
$\sigma(R^{stocks})/\sigma(R^b/\pi)$	1.7-8.9	4.1	4.9	3.8	4.1	7.2

Notes: Unconditional moments in U.S. data, 1950-2016, and in the model. $\sigma(x)$ and $\rho(x)$ denote the standard deviation and the autocorrelation, respectively, of variable x. $\rho(x, y)$ denotes the correlation of x and y. Δx denotes the growth rate of x. Appendix C.3 lists the composition of the aggregate variables. Stock market data by Robert Shiller (S&P 500). The model variants are as follows: (I): Two-Asset HANK with News; (II): Two-Asset HANK without News; (III): One-Asset HANK with News; (IV): One-Asset HANK without News; (V): Two-Asset HANK, only Noise
(*) denotes moments that were targeted during the calibration.

Table 4 shows the simulation results for various model variants and shock combinations, and compares them to the unconditional moments of the data. The main result is that the baseline variant, column (I), explains around two thirds of the fluctuation in the price-dividend ratio of the S&P 500. The comparison with columns (II) and (III) shows that news shocks and portfolio choice between liquid and illiquid assets are both important for explaining stock price fluctuations. Only the two-asset model allows for a time-varying illiquidity premium, which leads to larger fluctuations in the return to liquid assets and induces comovement between bond returns and stock returns (see the lower set of rows). In the one-asset economy, the correlation between stock returns and bond returns is close to zero, as surprise TFP and markup shocks cause surprise changes in dividend payments that are orthogonal to government bond returns. News shocks cause the illiquidity premium to fluctuate even more, but in a structured way: they add the boom-bust cycle. Thereby, news shocks can explain higher volatility of the price-dividend ratio, while at the same time generating some *momentum* $\rho(\Delta P/D)$, i.e., the autocorrelation in growth rates, which is a salient feature of the data, and causing comovement of the stock price cycle with aggregate consumption. The model predicts that investment and the stock market are more positively correlated than in the data. Adam and Merkel (2019) show that a subset of investment in fixed assets, namely non-residential investment and investment in non-residential structures, correlates more with the stock market.

Table 5: Campbell-Shiller decomposition in data and model

Source / Variant	Dividends	Discount rate	PD-ratio
Cochrane (2011)	0.11	1.01	0.11
Kuvshinov (2022)	0.55	0.45	-
Baseline	0.35	0.42	0.23
One-Asset	0.93	-0.00	0.07
No News	0.36	0.34	0.31
Only Noise	0.28	0.47	0.26

Notes: Variance shares of the Campbell-Shiller decomposition. For the model variants, I use the method by Cochrane (2011) to calculate the variance shares, with a time horizon of 15 years. "No News" and "Only Noise" are two-asset model variants.

However, since housing is the most important illiquid asset of the majority of households in the data, I cannot abstract from it in my quantitative model. Finally, as presented in column (V), noise shocks are almost equally successful in explaining stock price fluctuations. However, they imply that stock returns fluctuate 7 times more than returns on government bonds, close to the upper end of what is realistic, and fail to generate any momentum.

Next, I use the Campbell and Shiller (1988) decomposition to analyze the degree to which the model variants explain the salient feature of stock prices (and asset prices in general): return predictability. It is a log-linear approximation of the price-dividend ratio around its (proposed) stationary value, and is given by

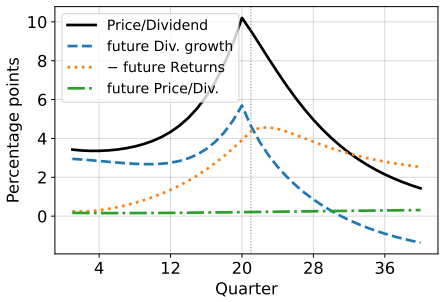
$$\log(q_t^{\Pi}/\Pi_t^F) = c + \mathbb{E}_t \sum_{j=0}^{\infty} \rho^j \left[\underbrace{\hat{\Pi}_{t+1+j}^F}_{\text{dividend growth news}} \qquad \underbrace{-r_{t+1+j}^L}_{\text{discount rate news}} \right], \tag{31}$$

where c and ρ are constants that are computed from long-run averages, and $r_t^L = R_t^L/\pi_t - 1$ is defined as the net real return on the liquid asset (where I assume that the no-arbitrage condition holds up to first order, i.e., r^L is also the expected net return on the stocks). The decomposition shows that the contemporaneous price-dividend ratio is determined by dividend growth news and negative discount rate news up to first order (in the formula with a finite horizon, a future price-dividend ratio also enters).

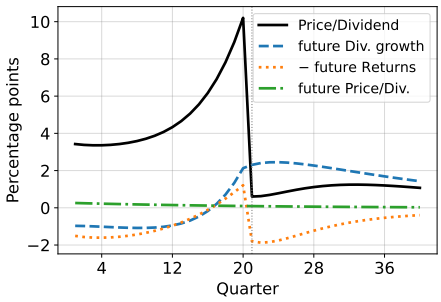
Table 5 shows the results of decomposing the variance of the log price-dividend ratio into the variances of the two news components and the future price-dividend ratio, in both data and the model variants. In the baseline model, discount rate news explain more than 40% of the variance in the price-dividend ratio. The results for the one-asset

Figure 14: Campbell-Shiller decomposition with countercyclical dividends (model)

a. Realized capital share increase



b. False expectations (ex post decomp.)



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21).
In b), the news is offset by a negative capital share surprise shock in quarter 21.

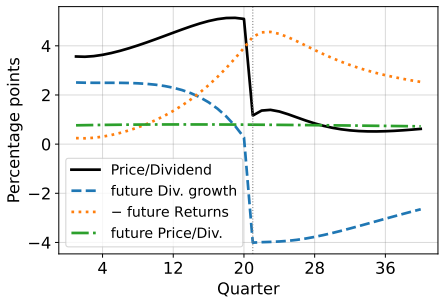
variant with news shocks and for the two-asset variant without news shocks show that the main cause of return predictability in the model is not news shocks, but the financial friction on the household side: the existence of wealthy, liquidity-constrained households, whose subjective discount factor varies with asset returns, is the key to generating time-varying discount rates. Naturally, the existence of news increases the predictive power of discount rate news, while noise shocks are able to generate an even higher importance for the discount rate component, at the expense of the predictive power of the "news".

Why is such a large share of stock price fluctuations in the model explained by expected future dividend growth? In panel a) of Figure 14, I plot the Campbell-Shiller decomposition as an impulse response to a capital share news shock. One can see that the price-dividend ratio correlates with future dividend growth. The reason is that dividends are countercyclical in the model (although profit smoothing mitigates this), so that a stock price boom that coincides with a business cycle boom automatically implies positive dividend growth news. In panel b), I plot the impulse response to a noise shock. Specifically, the contemporaneous price-dividend ratio, which up to the 21st quarter is driven by the wrong expectation of a capital share increase, is plotted together with the true future components that are known ex post. Now, of course, the Campbell-Shiller decomposition does not hold in the anticipation period, as the price is based on a wrong expectation. Indeed, as the real rate also falls after the disappointment of the news in the model, the future returns component can "rationalize" some of the excess price-dividend ratio relative to future dividend growth. Due to profit smoothing, future dividend growth fails to rationalize ex post the variation of the price-dividend ratio over the cycle.

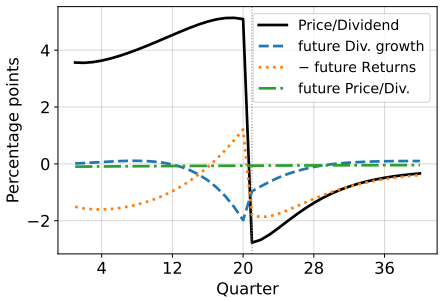
In order to illustrate the impact of the cyclicality of dividends on the results, I also

Figure 15: Campbell-Shiller decomposition with procyclical dividends

a. Realized capital share increase



b. False expectations (ex post decomp.)



Notes: Model impulse responses are to news about a temporary capital share increase in 5 years (quarter 21).
In b), the news is offset by a negative capital share surprise shock in quarter 21.
The hypothetical asset yields dividends $\omega^{\Pi} Y$.

compute the Campbell-Shiller decomposition for an alternative asset, where the dividend is simply given by a fraction of output (see Figure 15). For this type of stock, the future returns component explains the smooth increase of the price-dividend ratio in the anticipation phase, and its smooth decline in the subsequent bust phase. Future dividends instead explain the jumps in the price-dividend ratio, one at the onset of the news, and one at the onset of the productivity change. With constant returns (or discount rates), a forward-looking price would already incorporate the future expected decline in dividends at the onset of the news, and thus be mostly declining in the anticipation phase. But since the future dividends will also be discounted less as the demand for liquidity will rise, the price-dividend ratio rises in the anticipation phase. In Figure 15b, I show that if the capital share expectation is disappointed, the future returns (which increase quickly after the disappointment of the news, as the price level shoots up and then declines slowly) explain most of the subsequent lower stock price, while the future dividend component converges back to its steady state level.

6 Conclusion

What is the reason for the return predictability in the stock market? I propose a mechanism to explain this pervasive empirical pattern that hinges on incomplete markets and the existence of illiquid assets. I show in a quantitative business cycle model with time-separable preferences that the mechanism can account for a large part of the return predictability, as well as for other unconditional data moments of stock prices and the

business cycle. The main intuition behind the result is that the model accounts for the existence of wealthy marginal traders: wealthy households can be liquidity-constrained when they own mostly illiquid assets. In turn, asset income correlates with the business cycle, which induces a cyclicality of the stochastic discount factor of the marginal traders. Together with anticipation, these factors generate realistic stock price cycles.

The empirical evidence is in line with the proposed mechanism: first, returns on liquid and illiquid assets correlate with the stock market as expected. Second, I show using survey data that households who earn mostly capital income shift their wealth towards illiquid assets in stock price booms, and increase the liquidity of their portfolio in the subsequent stock price bust, as predicted by the model. In the model, a large part of stock price fluctuations stems from the portfolio choices of "rentiers": households who live mostly off the income of illiquid assets. This hypothesis should be investigated further, ideally using household-level data on both consumption and investment.

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A Challe-Ragot model

Preferences	
β	0.95
σ (risk preference)	1
c^*	10
Environment	
y^l	2
y^h	14
$\mathbb{P}(h \to l)$	10%
$\mathbb{P}(l \to h)$	90%
μ	1
k	50
$\bar{r}$	4%
$\bar{L}$	3
Steady state	
$\mathbb{P}(h)$	90%
$R-1$	3.53%
$\tilde{b}$	3.44

Table A.1: Calibration of the model parameters and steady state levels of variables

B Proof of Proposition 1

I show that

$$R_t^K - R_t^L \geq \beta(1-\lambda)\frac{\mathbb{E}_t^i\left[\gamma_{t+1}^i\right]}{q_t u'(c_t^i)} \tag{32}$$

for all households i that trade the illiquid asset in period t, where

$$\gamma_t^i := q_t u'(c_t^{i,n}) - \beta\,\mathbb{E}_t^i\, V_{t+1,k}(b_t^{i,n}, k_t^{i,n}, h_{t+1}) \geq 0, \tag{33}$$

and R^K and R^L are gross returns on illiquid and liquid assets, respectively, which, as the rest of the variable notation, follows the model description in Section 3. In addition to that notation, I add superscripts n or a also to variables that denote optimal choices of households that are ***n**ot* able to adjust their capital holdings that period, or are able to ***a**djust* their capital holdings that period, respectively. For the sake of readability, I omit the superscript i that identifies household i in the following proofs.

The proof boils down to deriving a "Basic Pricing Equation for illiquid assets" from the first-order optimality conditions of households that can adjust their capital holdings, and combining that with the standard Euler equation for liquid assets. The inequality accounts for households that are financially constrained. All expectations are only with respect to idiosyncratic uncertainty, as certainty equivalence holds with respect to aggregate variables.

Lemma 1. *For a household that is financially unconstrained in period $t-1$, the "Basic Pricing Equation for illiquid assets"*

$$1 = R_t^K \beta \, \mathbb{E}_{t-1} \left[\frac{u'(c_t^a) - (1-\lambda)\frac{\omega_t}{q_t + r_t}}{u'(c_{t-1}^a)} \right] \tag{34}$$

holds, where

$$\omega_t := (q_t + r_t)u'(c_t^a) - (r_t u'(c_t^n) + \beta \, \mathbb{E}_{t,h_t} V_{t+1,k}(b_t^n, k_t^n, h_{t+1})) \tag{35}$$

is the expected marginal benefit for the household of being able to adjust its illiquid portfolio holdings in period t.

Proof It holds that

$$\begin{aligned}
\mathbb{E}_{t-1,h_{t-1}} V_{t,k}(b,k,h_t) &= \mathbb{E}_{h_t|h_{t-1}} \left[\lambda V_{t,k}^a(b,k,h_t) + (1-\lambda) V_{t,k}^n(b,k,h_t) \right] && (36)\\
&= \mathbb{E}_{h_t|h_{t-1}} [\lambda(q_t + r_t)u'(c_t^a) + \\
&\quad (1-\lambda)(r_t u'(c_t^n) + \beta \, \mathbb{E}_{h_{t+1}|h_t} V_{t+1,k}(b_t^n, k, h_{t+1}))] && (37)\\
&= \mathbb{E}_{h_t|h_{t-1}} [-(1-\lambda)(\omega_t - r_t(u'(c_t^a) - u'(c_t^n))) + q_t u'(c_t^a) + \\
&\quad r_t(\lambda u'(c_t^a) + (1-\lambda)u'(c_t^n))] && (38)\\
&= \mathbb{E}_{h_t|h_{t-1}} [(q_t + r_t)u'(c_t^a) - (1-\lambda)\omega_t] && (39)
\end{aligned}$$

where the second equation follows from the envelope theorem.

Optimal illiquid asset choice in period $t-1$ requires that

$$q_{t-1} u'(c_{t-1}^a) = \beta \, \mathbb{E}_{t-1,h_{t-1}} V_{t,k}(b_{t-1}^a, k_{t-1}^a, h_t) \tag{40}$$

With $R_t^K = \frac{q_t + r_t}{q_{t-1}}$, substituting equation (39) into equation (40) gives the claim of the lemma. □

Note that the Euler equation with respect to the liquid asset requires

$$1 = R_t^L \beta \, \mathbb{E}_{t-1} \left[\frac{\lambda u'(c_t^a) + (1-\lambda) u'(c_t^n)}{u'(c_{t-1}^a)} \right] \tag{41}$$

for financially unconstrained households. Combining this and the equation from Lemma 1, one gets

$$\begin{aligned} \frac{1}{R_t^L} - \frac{1}{R_t^K} =& \beta(1-\lambda) \frac{\mathbb{E}_{t-1}\left[\frac{\omega_t}{q_t + r_t} - (u'(c_t^a) - u'(c_t^n)) \right]}{u'(c_{t-1}^a)} && (42) \\ =& \beta(1-\lambda) \frac{\mathbb{E}_{t-1}\left[u'(c_t^n) - \frac{r_t u'(c_t^n) + \beta \, \mathbb{E}_{t,h_t} V_{t+1,k}(b_t^n, k_t^n, h_{t+1})}{q_t + r_t} \right]}{u'(c_{t-1}^a)} && (43) \\ =& \beta(1-\lambda) \frac{\mathbb{E}_{t-1}\left[q_t u'(c_t^n) - \beta \, \mathbb{E}_{t,h_t} V_{t+1,k}(b_t^n, k_t^n, h_{t+1}) \right]}{(q_t + r_t) u'(c_{t-1}^a)} && (44) \end{aligned}$$

Multiplying by R_t^K yields

$$\frac{R_t^K - R_t^L}{R_t^L} = \beta(1-\lambda) \frac{\mathbb{E}_{t-1}[\gamma_t]}{q_{t-1} u'(c_{t-1}^a)} \tag{45}$$

As noted above, γ_t has the interpretation of the shadow price of the financial friction that prevents households from trading their illiquid assets. It is positive for households that cannot adjust their illiquid asset holdings, as optimality condition (40) is not fulfilled in that case (the case where the amount of the existing illiquid asset k remains optimal is a zero probability event). Since quarterly returns to liquid assets are close to one in most calibrations, the above equation approximately gives the quarterly (ex-ante) illiquidity premium $ILP := R_t^K - R_t^L$ that obtains in equilibrium, as determined by the optimization of households that are financially unconstrained in period $t-1$. For households at the borrowing constraint, the Euler equation (41) is violated in the direction of higher marginal utility today, which implies the claimed inequality. □

C Empirical evidence

C.1 Stock returns, capital rents, and business cycle variables

This section presents regressions of quarterly S&P 500 stock returns (data by Robert Shiller) on the growth of after-tax capital rents (Gomme et al., 2011) and other variables. The sample is split in two, periods where the trend of the S&P 500 return is rising, and periods where it is falling. The trends of the S&P stock return, inflation growth, and GDP

are computed using the Hodrick-Prescott filter with a smoothing parameter of 1600. All variables are standardized.

Findings:

- In periods of stock returns trending upwards (column (a) of Table C.1), stock returns are statistically significantly correlated with consumption growth (5% level), and weakly statistically significantly correlated with falling inflation and deviations of GDP from trend (10% level). There is no correlation with capital rents.
- In periods of stock returns trending downwards (columns (b) and (c)), stock returns are statistically significantly correlated with investment growth and dividend growth (1% level). Capital returns are weakly negatively correlated. However, without investment as regressor, capital returns become positively correlated with stock returns. This shows that investment and capital returns explain similar parts of the variance in downturns.

Table C.1: Regressions of quarterly stock returns, by the trend of the stock return

	(a) rising	(b) falling	(c) falling
Capital rent growth	−0.09 (0.11)	−0.03 (0.12)	0.13 (0.12)
Consumption growth	0.22** (0.11)	0.14 (0.09)	0.16* (0.09)
Investment growth	0.14 (0.10)	0.33*** (0.10)	–
Dividend growth	0.00 (0.09)	0.18*** (0.06)	0.23*** (0.07)
GDP deviation from trend	−0.18* (0.10)	−0.12 (0.09)	−0.16 (0.10)
Inflation trend rising	−0.14* (0.09)	−0.09 (0.07)	−0.06 (0.07)
Before 1980	0.05 (0.09)	−0.05 (0.07)	−0.02 (0.08)
Observations	135	140	140
Adj. R-squared	0.10	0.20	0.13

Notes: The dependent variable is the quarterly S&P 500 stock return. The columns are the subsets of observations where the trend of the stock return is (a) rising, (b) falling, and (c) falling, with investment growth left out. *Inflation trend rising* and *Before 1980* are indicator variables. All variables are standardized within the subset; there is no constant. Newey-West (one lag) standard errors in parentheses. Asterisks indicate significance at the 1% (***), 5% (**) or 10% (*) level.

C.2 Survey evidence

C.2.1 Data selection and definitions

This section presents more evidence about heterogeneous portfolio choice in the U.S. over time. I use the 20 years available in the SCF+ (Kuhn et al., 2020) between 1950 and

2019, and add the waves of 2022 and 2025 of the Survey of Consumer Finances, which I bring into the format of the SCF+. I split the sample into two subgroups of years: from 1950 to 1971, and from 1983 to 2025. The year 1977 is left out of the analysis in the main text for two reasons: first, the gap to sampled years before and after 1977 is 6 years, which is double the gap between most of the sampled years in the survey (3 years). Hence, computing differences between sampled years is less consistent when including the year 1977. Second, I find that 1977 is an outlier in terms of the main object of analysis in this paper, the group of households with high capital income: while the share of households with high capital income who are in the top 10% of the wealth distribution is 28% in the median year, it is only 13% in 1977. Conversely, the share of these households who are in the bottom half of the distribution is 28% in the median year, and 61% in 1977. The likely reasons for this discrepancy are issues with the imputations of total and capital income. Over all remaining years, N=90815 households are in the survey.

The first subgroup from 1950 to 1971 is from the older waves of the SCF, where capital income lumps together asset incomes from the following sources:

1. non-taxable investments (e.g., municipal bonds)
2. other interest
3. dividends
4. other business or investments, net rent, trusts, or royalties

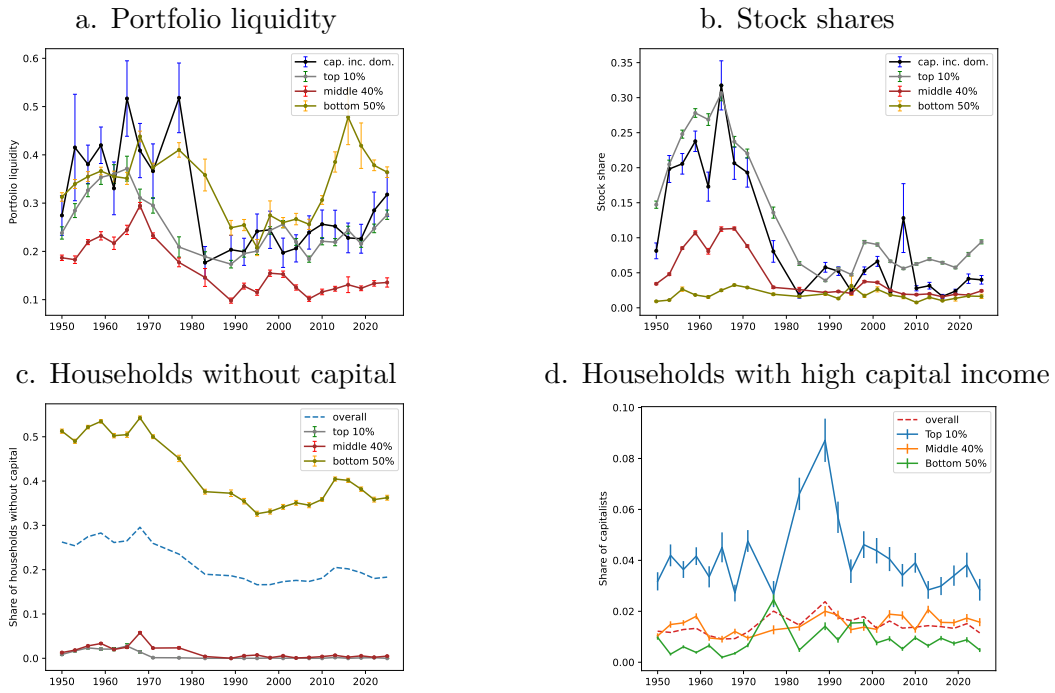
Since asset incomes number 2 and 3 likely stem from more liquid sources, namely treasury bonds and stocks, this definition of capital income does not fit the dichotomy between liquid and illiquid assets suggested by the analysis in the main text. Therefore, starting from year 1983 (the modern waves of the SCF), I sum up as a measure of capital income only income from the sources number 1 and 4.

Since the wave of 2004, source number 4 also contains the income from partnerships and S corporations, which the survey counted as business income before. Among the households with a high income from source number 4, about a third own a business before 2004, and two thirds to three quarters from 2007 on. To keep the group comparable over time, and because the wealth in a firm that a household manages itself is not illiquid for that household (see Section 5), I leave out of the high-capital-income group all households in which someone has an active management role in a business. The survey asks for this role from 1983 on. Between 10% and 12% of households have such a role in every wave.

In line with the quantitative model, I define as high capital income those households where capital income is at least 75% of their total income. In order to make this definition comparable across the old and modern waves of the SCF, where only the modern waves allow computing the model-consistent definition of capital income, I proceed in the following way: From the modern waves, I calculate the average share of asset income from sources number 1 and 4 among asset income from all sources, which equals 0.28. Then, I categorize households into the high-capital-income group in the *older* waves if at least 75% of their total income stems from the original capital income measure (with all sources), while for the *modern* waves, households' income must stem from sources number 1 and 4 at least at the rate of $75\% \cdot 0.28 = 21\%$ to be classified as high capital income. With this procedure, the average share of households with high capital income is similar in the older and the modern waves (1.1% and 1.5%).

C.2.2 Portfolio choice over time

Figure C.1: Heterogeneous portfolio choice over time



Notes: Survey evidence from SCF+ (Kuhn et al., 2020) and the SCF waves of 2022 and 2025. *Portfolio liquidity* is defined as the ratio of liquid assets to total wealth. *Stock shares* are defined as the ratio of stock wealth to total wealth. Households *without capital* are defined as households with zero illiquid wealth. Households with *high capital income* are households who earn a large share of capital income ($> 75\%$) compared to their overall income and who do not actively manage a business. Whiskers are 68%-confidence intervals.

Several secular trends are noticeable:

- For households in the top half of the wealth distribution, portfolio liquidity peaks in the 1960s, and has declined since then. Some of this development is due to a larger share of wealth held in stocks in the first half of the sample.

- For the bottom half of the wealth distribution, stocks are mostly irrelevant, and up to half of the households in that wealth category do not hold illiquid assets. The share of households without capital decreases from the 1970s on, and has increased again since the Great Recession. Also, the portfolio liquidity of the poorer households has increased markedly since 2008.

- The overall share of households with high capital income stays between 1% and 2.5% over the whole sample, without a trend. Their share within the richest decile peaks in the 1980s, at 7% to 9%, and lies between 3% and 5% otherwise.

C.3 Business cycle data

All series are available at quarterly frequency from the FRED database of the Federal Reserve Bank of St. Louis:

Output, Y: Sum of gross private domestic investment (GPDI), personal consumption expenditures for nondurable goods (PCND), durable goods (PCDG), and services (PCESV), and government consumption expenditures and gross investment (GCE) divided by the GDP deflator (GDPDEF) and the civilian noninstitutional population (CNP16OV).

Consumption, C: Sum of personal consumption expenditures for nondurable goods (PCND), durable goods (PCDG), and services (PCESV) divided by the GDP deflator (GDPDEF) and the civilian noninstitutional population (CNP16OV).

Investment, I: Gross private domestic investment (GPDI) divided by the GDP deflator (GDPDEF) and the civilian noninstitutional population (CNP16OV).

D Numerical implementation

The household's planning problem (10) includes the infinite-dimensional distribution Θ_t, so that the model has to be discretized. I solve the model with a first-order perturbation around its stationary equilibrium (Reiter, 2009, Bayer and Luetticke, 2020), where the household's problem is solved with the endogenous grid method (Carroll, 2006, Hintermaier and Koeniger, 2010). I deviate from the solution method of Bayer and Luetticke (2020) by using the endogenous gridpoint method for distributional dynamics of Bayer

et al. (2026) as the starting point. That method follows the cumulative distribution function instead of a histogram: it is moved forward with the household's policy evaluated at the endogenous gridpoints. For a given grid size, it is more accurate than the histogram method. Relative to those papers, two elements are new.

Conditional distributions instead of a fixed copula. The joint distribution is written as a sequence of conditional distributions: productivity, illiquid assets given productivity, and liquid assets given both. Besides the three marginal distributions, the state of the linearized model contains the deviations of these conditional distributions from their stationary values, on a coarse lattice of $20 \times 20 \times 10$ nodes. The dependence between liquid and illiquid wealth can therefore change over the cycle, while a fixed copula holds it constant. This matters for the question of this paper, because the mechanism runs through households who are rich in illiquid assets and at the same time poor in liquid assets.

Separate grids for policies and the distribution. The household's problem is solved on a finer grid (238×79 nodes for liquid and illiquid assets) than the grid on which the distribution is followed (80×40 nodes); productivity has $n_h = 10$ states on both. The policies are thus accurate around their kinks, at the borrowing limit and where the household starts to hold illiquid assets, without a larger state space of the linearized model.

The marginal value functions are, as in Bayer and Luetticke (2020), approximated around the stationary equilibrium by a sparse polynomial, using the discrete cosine transformation.

For the responses of groups of households (Figures 5, 8, 9 and 13), I simulate a panel of 500,000 households. Each quarter, a household acts according to the policy functions that the linearized model implies for the prices and the marginal values of that quarter. The policies are read off the endogenous gridpoints at the household's own cash at hand, so that households' asset holdings are not restricted to the grid. The panel is simulated three times with the same idiosyncratic shocks: in the stationary equilibrium, along the news shock that comes true, and along the news shock that is disappointed. The reported responses are differences to the panel in the stationary equilibrium. As the distribution is continuous, I count households within 0.05 of the borrowing limit as being at the limit.